

# The Cost of Following the Central Path

## Abstract

How much movement does following a central path add to the cost of reaching an accurate solution? We answer this question in specified barrier Hessian metrics on boxes, rectangular spectral-norm balls and Euclidean Jordan spectral intervals. For the standard logarithmic and log-determinant barriers, distance from the analytic center is exactly Euclidean in transformed spectral coordinates, and distance to the entire accurate target set is a one-multiplier convex allocation. For equal standard barrier scales and active objective rank  $r$ , the exact worst central-length-to-endpoint-distance ratio is  $\Gamma_r = [\sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2]^{1/2}$ , where  $\Gamma_r^2 = \frac{1}{4} \log r + O(1)$ . A certified scalar comparison gives a same-accuracy bound  $c_\star \Gamma_r$ , with  $1.37486 < c_\star < 1.38$ . An explicit dyadic sparse linear program has optimal primal movement  $\Theta(r\sqrt{\log r})$  and central-neighborhood movement  $\Theta(r \log r)$ , even for finite sequences with nonmonotone central labels. This separation persists for normalized scalar barriers, including the universal and entropic cube barriers, and uniformly for an explicit coupled family with optimal cube parameter. On the same linear program, returning any feasible small-gap primal-dual pair in the product barrier metric instead requires  $\Theta(r^{3/2})$  movement. Weighted exposed-rank bounds and concrete conic formulations clarify the role of auxiliary variables. The results count bounded local moves; they make no arithmetic- or oracle-complexity claim.

**Keywords:** interior-point methods; self-concordant barriers; central path; Riemannian distance; spectral optimization.

## Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                      | <b>2</b>  |
| 1.1      | Results and the mechanism behind them . . . . .                          | 3         |
| 1.2      | Relationship to prior work . . . . .                                     | 5         |
| 1.3      | Organization . . . . .                                                   | 6         |
| <b>2</b> | <b>Barrier geometry and the comparison problem</b>                       | <b>6</b>  |
| 2.1      | Definitions and local-step conversion . . . . .                          | 6         |
| <b>3</b> | <b>Exact distance and the closest accurate endpoint</b>                  | <b>8</b>  |
| 3.1      | Boxes, rectangular matrix balls, and Jordan intervals . . . . .          | 8         |
| 3.2      | Distance to an objective target set . . . . .                            | 11        |
| <b>4</b> | <b>Sharp comparison for the standard spectral barriers</b>               | <b>12</b> |
| 4.1      | The scalar velocity and the exact prefix constant . . . . .              | 13        |
| 4.2      | The exact scalar comparison with the closest accurate endpoint . . . . . | 14        |
| <b>5</b> | <b>Objective distributions and the amount of primal movement</b>         | <b>15</b> |
| 5.1      | Explicit lower certificates and their limits . . . . .                   | 17        |
| 5.2      | Two decay laws with matching arbitrary-path bounds . . . . .             | 18        |

|           |                                                                           |           |
|-----------|---------------------------------------------------------------------------|-----------|
| <b>6</b>  | <b>A rational family and finite central-neighborhood sequences</b>        | <b>18</b> |
| 6.1       | Arbitrary labels, actual accuracy and a growing metric tube . . . . .     | 19        |
| 6.2       | Unequal scales can produce a larger discrete separation . . . . .         | 21        |
| <b>7</b>  | <b>Dependence on the barrier</b>                                          | <b>21</b> |
| 7.1       | Every fixed normalized scalar barrier has the same sharp factor . . . . . | 22        |
| 7.2       | The same dyadic separation for every normalized profile . . . . .         | 23        |
| 7.3       | Universal and entropic cube barriers . . . . .                            | 24        |
| 7.4       | Transfer to trace-separable spectral metrics . . . . .                    | 25        |
| <b>8</b>  | <b>Coupled cube barriers with optimal parameter</b>                       | <b>26</b> |
| 8.1       | A lower bound from one regular facet collar . . . . .                     | 26        |
| 8.2       | Two explicit coupled barriers with the cube parameter . . . . .           | 27        |
| 8.3       | A uniform matching upper bound for the radial family . . . . .            | 28        |
| 8.4       | Radial coupling preserves the dyadic finite-sequence separation . . . . . | 30        |
| <b>9</b>  | <b>Primal movement and primal–dual completion</b>                         | <b>31</b> |
| 9.1       | Orthogonal speed allocation and the feasible gap set . . . . .            | 31        |
| 9.2       | An optimal cone barrier with little primal movement . . . . .             | 33        |
| 9.3       | Three movement scales on one sparse LP . . . . .                          | 34        |
| <b>10</b> | <b>What a change of formulation can preserve</b>                          | <b>35</b> |
| 10.1      | Exposed support rank and arbitrary auxiliary motion . . . . .             | 35        |
| 10.2      | A dimension-only consequence for arbitrary cone barriers . . . . .        | 37        |
| 10.3      | Grouped balls, norm trees, and PSD completion . . . . .                   | 38        |
| <b>11</b> | <b>Interpretation</b>                                                     | <b>39</b> |
| <b>A</b>  | <b>Rational certificates for the scalar dilation</b>                      | <b>39</b> |
| A.1       | An elasticity bound . . . . .                                             | 40        |
| A.2       | The strict upper certificate . . . . .                                    | 41        |
| A.3       | The strict lower certificate . . . . .                                    | 41        |
| A.4       | Every scalar-dilation maximizer lies in a certified interval . . . . .    | 42        |
| <b>B</b>  | <b>The scalar envelope and the nonmonotone construction</b>               | <b>42</b> |
| B.1       | A globally smooth construction without velocity monotonicity . . . . .    | 44        |
| <b>C</b>  | <b>Proofs for the concrete formulations</b>                               | <b>45</b> |
| C.1       | Grouped paraboloids and arbitrary PSD column packing . . . . .            | 45        |
| C.2       | Linked norm-tree centers and distance . . . . .                           | 46        |
| C.3       | Exact root-slice parameter . . . . .                                      | 46        |
| C.4       | Heterogeneous source counts and objective weights . . . . .               | 47        |

# 1 Introduction

A central path supplies both a route to an optimum and a scale for taking locally controlled steps. The route need not be shortest in the metric that controls those steps. To quantify the difference, one must specify what a competing route must accomplish: reach the same central point, reach any point with the same objective accuracy, or return a feasible primal–dual pair with a small gap.

These requirements lead to different answers even for one sparse linear program on an ordinary cube.

We study these answers in the real Hessian metric of a specified self-concordant barrier. A permitted move from  $x$  to  $y$  has bounded starting local norm,  $\|y - x\|_{F,x} \leq R$ , for fixed  $0 < R < 1$ . The minimum number of such moves is comparable to metric distance, up to integer rounding, with explicit constants depending only on  $R$ . This permits a direct comparison of centrality with the amount of movement actually needed for accuracy. The principal example has three scales:

$$\underbrace{r\sqrt{\log r}}_{\text{arbitrary primal movement}}, \quad \underbrace{r \log r}_{\text{central primal movement}}, \quad \underbrace{r^{3/2}}_{\text{feasible primal-dual movement}}.$$

The objective and tolerance are rational with polynomial ordinary encoding length. The first separation persists for iterates in a fixed central neighborhood, even when their reference central parameters backtrack. The second separation applies to every feasible terminal dual certificate in the specified product metric.

These are geometric comparisons, not operation counts. For example, a linear objective on a box can be optimized by reading its signs, and a spectral decomposition solves the corresponding support problem on a matrix ball. The examples isolate costs imposed by a movement rule. They do not imply a lower bound for algorithms permitted to bypass that rule, or an arithmetic or oracle advantage for a general optimization method.

## 1.1 Results and the mechanism behind them

The basic barrier is  $U(x) = \sum_i b(x_i)$  on  $(-1, 1)^r$ , where  $b(t) = -\log(1 - t^2)$ . Its scalar metric coordinate is  $\rho(t) = \int_0^t \sqrt{b''(u)} du$ . In the coordinates  $y_i = \rho(x_i)$ , distances are Euclidean and shortest paths are straight. A central path for a positive linear objective has coordinates  $y_i(s) = H(s - a_i)$ , where  $s$  is logarithmic central parameter,  $a_i$  is determined by the objective weight, and the common velocity  $H'$  increases from zero to one. Widely separated  $a_i$  activate successive groups of coordinates. The central curve bends through these successive directions, whereas the shortest route moves them together. [Figure 1](#) illustrates this mechanism.

Our main results develop this mechanism in four directions.

1. *Exact spectral geometry and the sharp rank factor.* For rectangular real or complex spectral-norm balls and spectral intervals in every Euclidean Jordan algebra, equipped with their standard log-determinant barriers, distance from the analytic center is exactly the Euclidean norm of the transformed singular values or eigenvalues. Leaving the objective's spectral frame cannot shorten this route. Distance to the entire accurate target set reduces to a strictly convex scalar allocation with one multiplier ([theorems 3.1 and 3.3](#)). For equal barrier scales and  $r$  active objective channels, every central subarc satisfies

$$L_{\text{CP}} \leq \Gamma_r d_F(\text{its endpoints}), \quad \Gamma_r = \left[ \sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2 \right]^{1/2}, \quad \Gamma_r^2 = \frac{1}{4} \log r + O(1). \quad (1.1)$$

The supremum of this ratio is exactly  $\Gamma_r$ . For the same objective accuracy,  $L_{\text{CP}}(\varepsilon) \leq c_\star \Gamma_r L_{\text{opt}}(\varepsilon)$ , where  $1.37486 < c_\star < 1.38$  is the sharp constant in a specified scalar KKT comparison, certified by rational arithmetic. Sharpness of the product  $c_\star \Gamma_r$  is not asserted ([theorems 4.1 and 4.3](#)). Unequal factor scales give an exact weighted prefix constant; its value depends on activation order and can have order  $\sqrt{r}$  when the scale range grows exponentially ([proposition 4.2](#)).

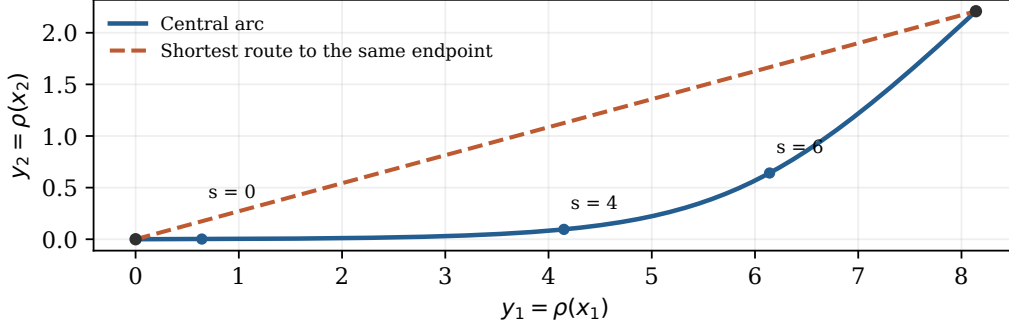


Figure 1: A two-coordinate example in the exact metric coordinates  $y_i = \rho(x_i)$ . The barrier is  $b(x_1) + b(x_2)$ , objective weights are  $(1, e^{-6})$ , and the terminal logarithmic parameter is  $s = 8$ . The central arc starts at the analytic-center limit  $s = -\infty$ . The straight segment is a globally shortest route to the same endpoint. The plot illustrates endpoint geometry; it does not display the closest point in an objective target set.

2. *A separation for actual finite sequences.* An explicit dyadic objective and tolerance on the cube give  $L_{\text{opt}} = \Theta(r\sqrt{\log r})$  and  $L_{\text{CP}} = \Theta(r \log r)$ , with ordinary input length  $\Theta(r^2)$  (theorem 6.1). A clipped progress function proves that every forward- $R$  sequence in a fixed metric tube around the central path needs  $\Omega(r \log r)$  moves to attain actual accuracy, with a matching  $O(r \log r)$  construction. The labels may be arbitrary and nonmonotone; no sampling assumption is imposed on the competing sequence. More generally, a diverging overhead survives tube radii  $\delta_r = o((\log r)^{1/3})$  (theorem 6.2). Fixed Newton-decrement neighborhoods of radius  $0 \leq \beta < 1/2$  satisfy this contract. The distinction between central arclength and finite sequences is necessary: an arclength lower bound alone does not exclude chords that skip a long part of a curve.
3. *Persistence beyond the standard separable barrier.* Every fixed scalar barrier satisfying the positive-branch normalization  $(f')^2 \leq f''$  has the same sharp endpoint factor  $\Gamma_r$ , a uniform same-accuracy comparison with constant  $C_{\text{sc}} < 2$ , and the same dyadic separation. This includes the universal and entropic cube barriers (theorem 7.1 and proposition 7.4). The scalar normalization has content: the unrestricted class admits ratios approaching  $\sqrt{r}$ , with a growing scalar parameter. Convex couplings with bounded first and second derivatives on one full facet collar retain a worst ratio at least  $\Gamma_r$  (theorem 8.1). We give dense and fully signed-permutation invariant coupled examples with exact, optimal cube parameter  $r$ . For the explicit radial family

$$F_{\lambda,c}(x) = U(x) - \lambda \log(c - \|x\|^2), \quad \lambda \geq 1, \quad c - r \geq 4\lambda,$$

we also prove a matching  $O(\Gamma_r)$  upper comparison and uniform persistence of the finite-sequence separation, even when  $\lambda, c$  vary with dimension (theorem 8.3 and corollary 8.4). Thus optimal parameter and substantial symmetry do not ensure a bounded centrality ratio. The matching upper theorem concerns this explicit family; the facet-collar theorem for general couplings is a lower bound.

4. *The additional cost of a dual completion.* Classical primal–dual geometry fixes total central speed at  $\sqrt{\nu}$  in logarithmic parameter. We calculate how objective activity divides this speed between primal and dual components. An optimal product-ball cone barrier permits much less primal

movement than its parameter alone would suggest, even with positive weights and a unique projected optimizer. Most directly, the same dyadic sparse LP has optimal primal movement  $\Theta(r\sqrt{\log r})$ , central primal movement  $\Theta(r \log r)$ , and movement to any feasible full small-gap point  $\Theta(r^{3/2})$  ([theorem 9.3](#)). The metric and endpoint contracts are stated separately in that theorem.

The supporting results identify when these comparisons survive additional structure. Effective objective rank gives explicit velocity and error laws, including geometric and polynomial spectral decay; a determinant rank certificate can miss an unbounded target distance ([section 5](#)). For changes of formulation, an exposed Jordan minor gives a weighted support-rank lower bound that allows all auxiliary fibers to move. A dimension cap on operational cone factors gives a full primal–dual lower bound for arbitrary logarithmically homogeneous barriers. Grouped balls, norm trees and arbitrary PSD column packing provide complete fixed-metric examples, including exact restricted parameters and heterogeneous objective scales ([section 10](#) and [appendix C](#)). These consequences distinguish which information comes from the objective, the barrier, and the lift.

## 1.2 Relationship to prior work

Nesterov and Todd [\[21\]](#) established the basic Riemannian geometry of self-concordant barriers, its connection to bounded local steps, and the product rule for distances. Their [Section 6.2](#) flattens a separable hypercube metric by integrating its scalar local norm. Their primal–dual theorem bounds central length by  $\sqrt{2}$  times endpoint distance in the metric of a logarithmically homogeneous cone barrier plus its conjugate. Their [Theorem 5.1\(c\)](#) and [Corollary 5.1](#) already extend the comparison to the entire feasible small-gap set, with possible affine infeasibility at intermediate points. We restate this classical result in [theorem 9.2](#) to make the contrast with primal shortcuts precise; the gap-set extension is not an original contribution.

Nesterov and Nemirovski [\[19\]](#) studied primal central paths versus geodesics. If  $D$  is the distance between two central points, their [Theorem 4.1](#) gives

$$L \leq \log 2 + \nu^{1/4} \sqrt{D(D + \log 12)}. \quad (1.2)$$

Their [Example 1.1](#) distinguishes a central endpoint from an objective target set, and their [Theorem 5.3](#) gives a target-set comparison under an explicit separation assumption. The present work addresses that established comparison problem on specified spectral domains: it computes the accurate-target benchmark and replaces the general parameter dependence by an exact active-rank constant for these metrics.

Worst-case central-path behavior and lower bounds for path-following methods have a substantial separate history. Deza, Nematollahi and Terlaky [\[9\]](#) study iteration lower bounds through Klee–Minty constructions. Allamigeon, Benchimol, Gaubert and Joswig [\[1\]](#) construct long-winding logarithmic central paths and prove exponential neighborhood iteration lower bounds. Allamigeon, Gaubert and Vandame [\[2\]](#) give exponential lower bounds for self-concordant-barrier path-following methods in their prescribed neighborhood model. Allamigeon, Dadush, Loho, Natura and Végh [\[3\]](#) compare an interior-point method with *straight-line complexity*, which counts affine segments in a wide central neighborhood, and obtain comparisons across self-concordant barriers. Dadush, Ma, Natura and Végh [\[8\]](#) improve wide-neighborhood guarantees expressed in terms of straight-line complexity and establish approximate instance optimality of trust-region path following in the narrow  $\ell_2$  neighborhood. Their affine-segment and algorithmic iteration resources differ from the fixed-radius local-Hessian moves counted here. Our dyadic example gives a separation between central-neighborhood movement and unrestricted target distance in the same metric on the same

simple domain; it is not a first demonstration of central-path inefficiency or a barrier-independent exponential iteration lower bound.

Several ingredients have direct classical antecedents. The prefix norm inequality is a finite weighted Lorentz-norm inequality [18]; the new comparison requires its sharp realization by translated central velocities. The matrix and Jordan Hessian calculations use spectral differentiation [4, 17]. Metric projections and conjugate Hessian isometries appear explicitly in Ohara, Ishi and Tsuchiya [22]. The universal and entropic constructions and their exact dimension parameters are due to earlier work [5, 7, 15, 20]; we derive their centrality consequences. Parameter-preserving diagonal quadratic regularization on boxes is also prior work [6]. The coupled examples here supplement that construction literature with exact-parameter path comparisons.

Geodesic updates in symmetric-cone algorithms [24] and interior-point methods on Riemannian base manifolds [14] concern complementary algorithmic settings. The real Hessian metric on a bounded spectral ball used here is distinct from the affine-invariant metric on a symmetric cone and from complex Bergman metrics. The formulas are proved for the stated metric, including complex matrix spaces viewed as real vector spaces.

To the best of our knowledge, the exact active-rank supremum (1.1) for these central paths, its combination with the explicit accurate-target allocation, the dyadic finite-sequence separation with arbitrary labels, the uniform comparison for the stated radial coupled family, and the same-instance three-scale completion comparison have not appeared in the cited literature. These are the specific original claims supported by our proofs and literature review. The general geometric framework, canonical barriers, and classical primal–dual gap-set comparison are antecedents of these results.

### 1.3 Organization

Section 2 fixes the metric and local-step conventions. Sections 3 and 4 establish the exact benchmark and sharp comparison; sections 5 and 6 develop objective profiles and finite sequences. Sections 7 and 8 treat scalar and coupled changes of barrier. Sections 9 and 10 separate dual completion from formulation effects. The appendices give the rational scalar certificate, the normalized profile envelope and nonmonotone construction, and the complete proofs for the concrete formulations. The figures are reproducible numerical illustrations of analytically proved results.

## 2 Barrier geometry and the comparison problem

The quantity studied here is the amount of local movement required to reach an accurate solution. There are three distinct comparisons: a central arc versus a shortest path to the same endpoint; a central arc versus a shortest path to any sufficiently accurate endpoint; and primal movement versus movement of a feasible primal–dual pair. The choice of endpoint set and metric is part of each statement.

### 2.1 Definitions and local-step conversion

Let  $\Omega$  be an open convex subset of a finite-dimensional real inner product space  $E$ . A  $C^3$  function  $F : \Omega \rightarrow \mathbb{R}$  with positive definite Hessian defines

$$\|h\|_{F,x}^2 = D^2F(x)[h, h], \quad \text{len}_F(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|_{F,\gamma(t)} dt, \quad d_F(x, y) = \inf_{\gamma: x \rightsquigarrow y} \text{len}_F(\gamma). \quad (2.1)$$

The infimum is over absolutely continuous curves in  $\Omega$ . For a convex affine slice all norms are restricted to its tangent space, and curves remain in that slice. Complex matrix spaces are always treated as real spaces, with inner product  $\text{Re tr}(X^*Y)$ .

We call  $F$  a standard self-concordant barrier if it tends to  $+\infty$  at every finite boundary point and satisfies

$$|D^3F(x)[h, h, h]| \leq 2\|h\|_{F,x}^3. \quad (2.2)$$

Its barrier parameter is any  $\nu$  satisfying  $|DF(x)[h]| \leq \sqrt{\nu}\|h\|_{F,x}$  for all  $x, h$ . The adjective “standard” here fixes the constant 2 in (2.2); it does not prescribe a particular barrier. The polarized self-concordance inequality is

$$|D^3F(x)[u, h, h]| \leq 2\|u\|_{F,x}\|h\|_{F,x}^2. \quad (2.3)$$

We suppress  $F$  in local norms when it is fixed.

**Lemma 2.1** (Distance and a forward chord). *For a standard self-concordant barrier,*

$$\log(1 + \|y - x\|_x) \leq d_F(x, y). \quad (2.4)$$

*If  $q = \|y - x\|_x < 1$ , the straight segment from  $x$  to  $y$  has length at most  $-\log(1 - q)$ . In particular, for  $0 < R < 1$ ,*

$$\|y - x\|_x \leq R \implies d_F(x, y) \leq -\log(1 - R), \quad d_F(x, y) \leq \log(1 + R) \implies \|y - x\|_x \leq R. \quad (2.5)$$

*Proof.* Along any curve  $\gamma$  starting at  $x$ , let  $\ell(t) = \int_0^t \|\dot{\gamma}(u)\|_{\gamma(u)} du$ . For a fixed vector  $h \neq 0$ , (2.3) gives  $|\frac{d}{dt} \log\|h\|_{\gamma(t)}| \leq \dot{\ell}(t)$  almost everywhere. Thus  $\|h\|_x \leq e^{\ell(t)}\|h\|_{\gamma(t)}$  for every  $h$ . The triangle inequality in the fixed norm at  $x$  now yields

$$\|y - x\|_x \leq \int_0^1 \|\dot{\gamma}(t)\|_x dt \leq \int_0^1 e^{\ell(t)} \dot{\ell}(t) dt = e^{\text{len}_F(\gamma)} - 1.$$

Taking the infimum proves (2.4). For the segment  $x + th$ ,  $h = y - x$ , set  $a(t) = \|h\|_{x+th}$ . Self-concordance gives  $|a'(t)| \leq a(t)^2$  and hence  $a(t) \leq q/(1 - tq)$  while  $tq < 1$ . Integration from 0 to 1 gives the second assertion. The segment lies in  $\Omega$  by convexity.  $\square$

These are the local geometric estimates behind the short-step counting results of Nesterov and Todd [21, Section 3]. For a target  $A \subset \Omega$  and starting point  $x_0$ , put  $D_F(x_0, A) = \inf_{x \in A} d_F(x_0, x)$ . A forward- $R$  sequence is a finite feasible sequence  $x_0, x_1, \dots, x_T$  such that  $\|x_{j+1} - x_j\|_{x_j} \leq R$ .

**Corollary 2.2** (Optimal bounded movement). *Let  $F$  be a standard self-concordant barrier and  $0 < R < 1$ . Every forward- $R$  sequence from  $x_0$  to  $A$  satisfies*

$$T \geq \frac{D_F(x_0, A)}{-\log(1 - R)}. \quad (2.6)$$

*If a shortest curve to  $A$  exists, there is such a sequence with*

$$T \leq \left\lceil \frac{D_F(x_0, A)}{\log(1 + R)} \right\rceil. \quad (2.7)$$

*Proof.* Concatenate the straight chords and apply lemma 2.1 to prove the lower bound. For the upper bound, partition a shortest curve into arcs of length at most  $\log(1 + R)$  and apply (2.4) to their endpoints. If the distance is zero and attained, no move is needed.  $\square$

For a bounded domain and nonzero  $c \in E$ , define

$$h_\Omega(c) = \sup_{x \in \Omega} \langle c, x \rangle, \quad g_c(x) = h_\Omega(c) - \langle c, x \rangle, \quad A_\varepsilon = \{x \in \Omega : g_c(x) \leq \varepsilon\}. \quad (2.8)$$

When  $F$  is strictly convex and a barrier, the central point is

$$x(\eta) = \arg \min_{x \in \Omega} \{F(x) - \eta \langle c, x \rangle\}, \quad \eta \geq 0. \quad (2.9)$$

Its value at zero is the analytic center  $x_0$ . We write  $L_{\text{opt}}(\varepsilon) = D_F(x_0, A_\varepsilon)$ ,  $L_{\text{CP}}(\eta) = \text{len}_F(x|_{[0, \eta]})$ , and  $L_{\text{CP}}(\varepsilon) = L_{\text{CP}}(\eta_\varepsilon)$  when  $\eta_\varepsilon$  is the first parameter for which  $g_c(x(\eta_\varepsilon)) = \varepsilon$ . The argument of  $L_{\text{CP}}$  is specified whenever ambiguity is possible.

### 3 Exact distance and the closest accurate endpoint

A single scalar coordinate describes all the metrics in this section. Let

$$b(t) = -\log(1 - t^2), \quad \rho(t) = \int_0^t \frac{\sqrt{2(1 + u^2)}}{1 - u^2} du, \quad -1 < t < 1. \quad (3.1)$$

Then  $\rho$  is odd, strictly increasing, maps  $(-1, 1)$  onto  $\mathbb{R}$ , and  $\rho'(t)^2 = b''(t)$ . For  $0 \leq t < 1$ ,

$$\log \frac{1}{1 - t} \leq \rho(t) \leq \sqrt{2} \log \frac{1}{1 - t}. \quad (3.2)$$

Indeed, after multiplying the corresponding derivative inequalities by  $1 - t$ , these bounds reduce to  $1 \leq \sqrt{2(1 + t^2)}/(1 + t) \leq \sqrt{2}$ .

#### 3.1 Boxes, rectangular matrix balls, and Jordan intervals

For a box  $(-1, 1)^n$  with  $F(x) = \sum_i \alpha_i b(x_i)$ ,  $\alpha_i > 0$ , integration of the diagonal metric gives

$$d_F(x, y)^2 = \sum_i \alpha_i (\rho(x_i) - \rho(y_i))^2. \quad (3.3)$$

The transformed straight segment stays in the box and attains this distance. This scalar product construction is classical; the point of the next theorem is that leaving a spectral frame cannot shorten the route from the center.

For  $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$  and  $p \leq q$ , put

$$\mathcal{B}_{p,q}^{\mathbb{F}} = \{X \in \mathbb{F}^{p \times q} : \|X\|_{\text{op}} < 1\}, \quad \Phi_\alpha(X) = -\alpha \log \det(I_p - XX^*), \quad \alpha > 0. \quad (3.4)$$

Write  $\sigma^\downarrow(X)$  for its  $p$  singular values, in decreasing order, including zeros.

We also use a Euclidean Jordan algebra  $V$ , of rank  $n$ , with unit  $e$ , cone of squares  $K$ , Jordan product  $x \circ y$ , and trace inner product  $\langle x, y \rangle = \text{tr}(x \circ y)$ . A Jordan frame  $(e_i)_{i=1}^n$  consists of orthogonal primitive idempotents with sum  $e$ . The spectral theorem gives  $x = \sum_i \lambda_i(x) e_i$  and  $\det x = \prod_i \lambda_i(x)$ . A primitive idempotent is a nonzero element  $u$  with  $u \circ u = u$  that cannot be written as the sum of two nonzero orthogonal idempotents. The spectral interval and its barrier are

$$\mathcal{I}(V) = \{x : -e \prec_K x \prec_K e\}, \quad \Phi_\alpha(x) = -\alpha \log \det(e - x^2) = \alpha \sum_{i=1}^n b(\lambda_i(x)). \quad (3.5)$$

Here  $x^2 = x \circ x$  and  $u \prec_K v$  means  $v - u \in \text{int } K$ . The examples include real, complex, and quaternionic Hermitian matrices, Lorentz algebras, their direct products, and the exceptional Albert algebra. Rank, rather than the real dimension of  $V$ , counts the spectral coordinates.



**Theorem 3.1** (Exact spectral distance). *For every  $\alpha > 0$ , the following identities hold:*

$$d_{\Phi_\alpha}(0, X)^2 = \alpha \sum_{i=1}^p \rho(\sigma_i(X))^2, \quad X \in \mathcal{B}_{p,q}^{\mathbb{F}}, \quad (3.6)$$

$$d_{\Phi_\alpha}(0, x)^2 = \alpha \sum_{i=1}^n \rho(\lambda_i(x))^2, \quad x \in \mathcal{I}(V). \quad (3.7)$$

More generally,

$$d_{\Phi_\alpha}(X, Y) \geq \sqrt{\alpha} \|\rho(\sigma^\downarrow(X)) - \rho(\sigma^\downarrow(Y))\|_2, \quad (3.8)$$

$$d_{\Phi_\alpha}(x, y) \geq \sqrt{\alpha} \|\rho(\lambda^\downarrow(x)) - \rho(\lambda^\downarrow(y))\|_2. \quad (3.9)$$

Here  $\rho$  acts coordinatewise and the Jordan eigenvalues are ordered by signed value. Equality holds in each contraction when the two endpoints have a common ordered spectral frame. The minimizing path is linear in the  $\rho$  coordinates of that frame.

*Proof.* First consider matrices. Introduce the Hermitian dilation and a scalar function

$$S(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \quad g(t) = -\frac{1}{2} \log(1 - t^2).$$

The eigenvalues of  $S(X)$  are  $\pm\sigma_i(X)$  and  $q - p$  zeros, so  $\Phi_\alpha(X) = \alpha \operatorname{tr} g(S(X))$ . In an eigenbasis of a Hermitian matrix  $S$  with eigenvalues  $\lambda_a$ , the spectral-trace Hessian is

$$D^2 \operatorname{tr} g(S)[H, H] = \sum_a g''(\lambda_a) |H_{aa}|^2 + 2 \sum_{a < b} \frac{g'(\lambda_a) - g'(\lambda_b)}{\lambda_a - \lambda_b} |H_{ab}|^2. \quad (3.10)$$

The quotient at equal eigenvalues means  $g''(\lambda_a)$ . For completeness, the gradient of  $\operatorname{tr} g(S)$  is  $g'(S)$ . Differentiating a polynomial in  $S$  shows that the  $(a, b)$  entry of  $Dg'(S)[H]$  is the displayed divided difference times  $H_{ab}$ . The convergent power series of  $g$  on  $(-1, 1)$  justifies the same calculation here, uniformly on compact spectral subintervals. Taking its inner product with  $H$  gives (3.10), over the real and complex fields alike. This is the trace-separable case of the spectral Hessian formula [17].

All divided differences in (3.10) are positive. The ordered eigenvalues of an absolutely continuous Hermitian path are absolutely continuous; almost everywhere their derivatives are the eigenvalues of the derivative compressed to each repeated eigenspace. Thus one can choose the eigenbasis within each such space to identify the diagonal contributions with these derivatives. Applying the formula to  $S(X(t))$  gives

$$\|\dot{X}(t)\|_{\Phi_\alpha, X(t)}^2 \geq \alpha \sum_i b''(\sigma_i(X(t))) \dot{\sigma}_i(X(t))^2 = \alpha \left\| \frac{d}{dt} \rho(\sigma(X(t))) \right\|_2^2 \quad \text{a.e.} \quad (3.11)$$

The factor is  $b''$ , because each pair contributes  $2g'' = b''$ . Singular values are Lipschitz functions of the matrix, and at zero or repeated singular values the same almost-everywhere rule follows from the dilation. Integration proves (3.8). For a fixed singular value decomposition  $X = U \operatorname{diag}(\sigma_i) V^*$ , the path

$$X(t) = U \operatorname{diag}(\rho^{-1}(t\rho(\sigma_i))) V^*, \quad 0 \leq t \leq 1, \quad (3.12)$$

has constant speed equal to the right side of (3.6) to the power  $1/2$ . Indeed, restricting  $\Phi_\alpha$  to this frame gives  $\alpha \sum_i b(x_i)$ . The lower bound therefore is attained. Interpolation between two coordered transformed spectra gives the other equality assertion; order is preserved under this interpolation.

Here are the details for Jordan algebras. Relative to a Jordan frame, the Peirce spaces are  $V_{ii} = \mathbb{R}e_i$  and, for  $i < j$ ,

$$V_{ij} = \{h : e_i \circ h = e_j \circ h = \tfrac{1}{2}h, e_k \circ h = 0 \text{ for } k \notin \{i, j\}\}.$$

At  $x = \sum_i \lambda_i e_i$ , write the orthogonal Peirce decomposition of a vector  $h$  as  $h = \sum_i h_i e_i + \sum_{i < j} h_{ij}$ , where  $h_{ij} \in V_{ij}$  and  $\|e_i\| = 1$ . The Hessian is

$$D^2\Phi_\alpha(x)[h, h] = \alpha \sum_i b''(\lambda_i) h_i^2 + \alpha \sum_{i < j} \frac{b'(\lambda_i) - b'(\lambda_j)}{\lambda_i - \lambda_j} \|h_{ij}\|^2. \quad (3.13)$$

One way to verify this formula without restricting the algebra type is to differentiate the gradient  $\alpha((e - x)^{-1} - (e + x)^{-1})$ . To compute the derivative of inversion at a positive diagonal element  $z = \sum_i z_i e_i$ , let  $L_z h = z \circ h$  and differentiate  $z \circ z^{-1} = e$ . This gives  $L_z D(z^{-1})[h] = -L_{z^{-1}} h$ . The operator  $L_z$  is positive definite: it acts by  $z_i$  on  $V_{ii}$  and by  $(z_i + z_j)/2$  on  $V_{ij}$ . On the same spaces,  $L_{z^{-1}}$  acts by  $z_i^{-1}$  and  $(z_i^{-1} + z_j^{-1})/2$ , respectively. Solving the differentiated identity therefore gives the inversion coefficients  $-z_i^{-2}$  and  $-(z_i z_j)^{-1}$ . The resulting off-diagonal coefficient is

$$\frac{1}{(1 - \lambda_i)(1 - \lambda_j)} + \frac{1}{(1 + \lambda_i)(1 + \lambda_j)} = \frac{b'(\lambda_i) - b'(\lambda_j)}{\lambda_i - \lambda_j}.$$

This also gives its continuous value at a repeated eigenvalue. Every coefficient is positive. Jordan eigenvalues obey the same local Lipschitz and repeated-eigenspace derivative rule: within the Peirce algebra for a repeated eigenvalue, diagonalize the corresponding component of  $h$ . Equivalently, this follows from the spectral variational principle applied to  $x + th$ . Hence, along an absolutely continuous curve,

$$\|\dot{x}\|_{\Phi_\alpha, x}^2 \geq \alpha \sum_i b''(\lambda_i(x)) \dot{\lambda}_i(x)^2.$$

Integrate and interpolate in a fixed Jordan frame exactly as above. Only the Jordan spectral theorem and Peirce rules have been used, so the argument includes the exceptional algebra.  $\square$

The scale condition in the theorem is only  $\alpha > 0$ , since distance uses the Hessian. The local-step conclusions need self-concordance.

**Proposition 3.2** (Self-concordance and parameter). *If  $\alpha \geq 1$ , the barriers in (3.4) and (3.5) are standard self-concordant barriers. Their least gradient parameters are respectively  $\alpha p$  and  $\alpha n$ . A product with scales  $\alpha_j \geq 1$  has least gradient parameter equal to the sum of these parameters.*

*Proof.* The matrix barrier at scale one is the restriction of the positive Hermitian log-determinant barrier to  $\begin{pmatrix} I_p & X \\ X^* & I_q \end{pmatrix}$ . The Jordan barrier is the restriction of  $-\log \det u - \log \det v$  to  $(u, v) = (e - x, e + x)$ . The log-determinant is self-concordant: after its Hessian isometry at an interior point, the second and third directional derivatives are  $\sum_i \xi_i^2$  and  $-2 \sum_i \xi_i^3$ , so  $|\sum_i \xi_i^3| \leq (\sum_i \xi_i^2)^{3/2}$ . Affine restriction and addition preserve self-concordance, as does multiplication by  $\alpha \geq 1$ . Boundary divergence follows from the vanishing determinant.

The gradient has no component outside the spectral frame, and the Hessian has no mixed term between that frame and its orthogonal complement. Thus its squared dual norm is

$$\|\nabla \Phi_\alpha\|_{\Phi_\alpha, *}^2 = \alpha \sum_i \frac{b'(t_i)^2}{b''(t_i)} = \alpha \sum_i \frac{2t_i^2}{1 + t_i^2} < \alpha n_0, \quad (3.14)$$

where  $n_0 = p$  and  $t_i = \sigma_i$  for a matrix ball, and  $n_0 = n$  and  $t_i = \lambda_i$  for a Jordan interval. In the matrix case this block structure follows either from (3.10) or by differentiating the invariant trace function in singular coordinates. Letting all  $t_i$  tend to 1 proves that the supremum is  $\alpha n_0$ . For products, squared gradient dual norms add, and the limits can be approached simultaneously.  $\square$

### 3.2 Distance to an objective target set

Consider any finite product of the domains just defined. Each factor has its own scale  $\alpha_j > 0$ . A nonzero support objective has active weights  $w_i > 0$ : the positive singular values of each matrix objective, or the absolute values of the nonzero Jordan eigenvalues of each Jordan objective. Pool all these weights, indexed by  $1 \leq i \leq r$ , and attach to each its factor's scale  $\alpha_i$ . Within a factor all scales are therefore equal. For a scalar box this simply means  $w_i = |c_i|$  at the nonzero coordinates. Let  $W_0 = \sum_i w_i$ ; the support value is  $W_0$  and the analytic center is zero.

**Theorem 3.3** (The closest accurate endpoint). *For  $0 < \varepsilon < W_0$ , the distance to the entire objective target set is*

$$L_{\text{opt}}(\varepsilon)^2 = \min \left\{ \sum_{i=1}^r \alpha_i \rho(x_i)^2 : 0 \leq x_i < 1, \quad \sum_{i=1}^r w_i(1 - x_i) \leq \varepsilon \right\}. \quad (3.15)$$

*The scalar allocation problem has a unique minimizer, all coordinates lie in  $(0, 1)$ , and its constraint is active. There is a unique  $\lambda > 0$  such that*

$$2\alpha_i \rho(x_i) \rho'(x_i) = \lambda w_i \quad (1 \leq i \leq r), \quad \sum_i w_i(1 - x_i) = \varepsilon. \quad (3.16)$$

*An aligned endpoint with these coordinates and all inactive coordinates zero attains the minimum distance in the original domain. The path linear in its transformed spectral coordinates is globally shortest to the target set.*

*Proof.* Distances in a product metric add in squares. To see this directly, the length of a product curve is the integral of the Euclidean norm of its component speeds, which is at least the Euclidean norm of the component lengths. Constant-speed shortest component curves attain equality; such curves from zero were constructed in [theorem 3.1](#).

For a matrix factor, von Neumann's trace inequality gives  $\langle C, X \rangle \leq \sum_i w_i \sigma_i(X)$ , with weights and singular values ordered. Thus replacing an endpoint by one in the objective's singular frame can only improve its objective and leaves its distance unchanged. Inactive singular values can then be set to zero, reducing the distance. For a Jordan factor, if  $(e_i)$  and  $(f_j)$  are frames of  $c$  and  $x$ , the entries  $m_{ij} = \langle e_i, f_j \rangle$  are nonnegative and their row and column sums are one. Therefore

$$\langle c, x \rangle = \sum_{i,j} \mu_i \lambda_j m_{ij} \leq \sum_i |\mu_i|^\downarrow |\lambda_i|^\downarrow.$$

The last inequality is the rearrangement inequality for a doubly stochastic matrix. Assigning magnitudes to the objective frame in this order and using the signs of  $\mu_i$  attains the bound, at the same radial distance. Zero-weight coordinates can again be removed. Conversely every scalar vector in (3.15) gives an aligned feasible endpoint with precisely that objective error and radial distance. This proves the reduction.

For  $x \geq 0$ ,  $\rho' > 0$ ,  $\rho'' \geq 0$ , and  $(\rho^2)'' = 2((\rho')^2 + \rho\rho'') > 0$ . The objective is therefore strictly convex. It tends to infinity if any coordinate tends to 1. A nonempty bounded sublevel set is compact in  $[0, 1)^r$ , so a minimizer exists and is unique. Its accuracy constraint must be active: otherwise reduce one positive coordinate slightly. Slater's condition holds by taking every coordinate sufficiently close to 1. The KKT multiplier is positive, since a zero multiplier would force every coordinate to be zero, which violates  $\varepsilon < W_0$ . At  $x_i = 0$  the objective's derivative is zero, whereas the accuracy constraint contributes  $-\lambda w_i < 0$ ; stationarity with a nonnegative multiplier for  $-x_i \leq 0$  is then impossible. Every coordinate is positive, and stationarity gives (3.16).

The function  $(\rho^2)'$  increases continuously from zero to infinity. For a fixed  $\lambda > 0$ , each equation in (3.16) thus has one solution. Their weighted residual decreases continuously and strictly from  $W_0$  to zero as  $\lambda$  increases from zero to infinity. This proves uniqueness of  $\lambda$  and supplies a scalar search. The minimizing path follows from theorem 3.1 and the product rule. Within each factor the minimizer is ordered with its weights because its scales agree, so the common-frame construction applies without an ordering issue.  $\square$

The uniqueness assertion concerns the scalar allocation. It is not needed to assert uniqueness of all ambient matrix representations at repeated objective eigenvalues. If  $\varepsilon \geq W_0$ , the analytic center has zero cost. For  $\varepsilon = 0$ , no strictly interior point is accurate, and  $L_{\text{opt}}(\varepsilon) \rightarrow \infty$  as  $\varepsilon \downarrow 0$ .

**Corollary 3.4** (Logarithmic allocation and bounded moves). *Let  $0 < \varepsilon < W_0$ . Define*

$$D_{\log}(\varepsilon)^2 = \min \left\{ \sum_i \alpha_i s_i^2 : s_i \geq 0, \quad \sum_i w_i e^{-s_i} \leq \varepsilon \right\}. \quad (3.17)$$

*Then*

$$D_{\log}(\varepsilon) \leq L_{\text{opt}}(\varepsilon) \leq \sqrt{2} D_{\log}(\varepsilon). \quad (3.18)$$

*Its unique optimizer satisfies*

$$s_i = \mathcal{W}(\gamma w_i / \alpha_i), \quad \sum_i \alpha_i \mathcal{W}(\gamma w_i / \alpha_i) = \gamma \varepsilon, \quad \gamma > 0, \quad (3.19)$$

*where  $\mathcal{W}$  is the nonnegative inverse of  $s \mapsto s e^s$ . For  $0 < R < 1$ , if all scales are at least one, the minimum number  $T_R^*(\varepsilon)$  of feasible forward- $R$  moves from zero to any accurate endpoint obeys*

$$\frac{L_{\text{opt}}(\varepsilon)}{-\log(1-R)} \leq T_R^*(\varepsilon) \leq \left\lceil \frac{L_{\text{opt}}(\varepsilon)}{\log(1+R)} \right\rceil. \quad (3.20)$$

*Proof.* Set  $s_i = -\log(1 - x_i)$  and apply (3.2) to every feasible allocation. Strict convexity and the KKT conditions for (3.17) give  $\alpha_i s_i = \gamma w_i e^{-s_i}$  with all  $s_i > 0$ . Inverting and summing these identities gives (3.19). Uniqueness follows, for example, by observing that  $\sum_i w_i \exp[-\mathcal{W}(\gamma w_i / \alpha_i)]$  decreases strictly from  $W_0$  to zero; the zero root of the multiplied equation in (3.19) is excluded by  $\gamma > 0$ . Finally, proposition 3.2 and corollary 2.2 apply, and the shortest curve was constructed in theorem 3.3.  $\square$

The construction is explicit once the objective frame is available. Invert  $(\rho^2)'$  coordinatewise and use doubling and bisection for the positive multiplier. An upper bracket whose residual is at most the requested tolerance gives an accurate endpoint. Along the resulting path, checkpoints at arc intervals  $\log(1+R)$  satisfy the forward local-norm restriction. Finite-precision evaluation must preserve these two inequalities, for instance by using strict accuracy and step-radius margins. We make no arithmetic-complexity claim for this scalar search or for obtaining the spectral frame.

## 4 Sharp comparison for the standard spectral barriers

The reduction in theorem 3.3 applies to all products considered there. An active channel means a positive singular value of a matrix objective, or the absolute value of a nonzero Jordan eigenvalue. Denote these weights by  $w_i > 0$  and their factor scales by  $\alpha_i > 0$ ,  $i = 1, \dots, r$ . Signs and spectral frames are fixed by the objective; zero channels remain zero. The metric assertions in this section require only positive scales. Statements about standard self-concordance and bounded local moves additionally require  $\alpha_i \geq 1$ .

#### 4.1 The scalar velocity and the exact prefix constant

Put

$$q(z) = \frac{z}{1 + \sqrt{1 + z^2}}, \quad H(s) = \rho(q(e^s)), \quad v(s) = H'(s) = \sqrt{1 - \frac{1}{\sqrt{1 + e^{2s}}}}. \quad (4.1)$$

Here  $H$  is a strictly increasing bijection from  $\mathbb{R}$  to  $(0, \infty)$ . The central coordinates, in the objective frame, are

$$x_i(\eta) = q(\eta w_i / \alpha_i), \quad a_i = \log(\alpha_i / w_i), \quad y_i(s) = \rho(x_i(e^s)) = H(s - a_i). \quad (4.2)$$

For a matrix factor of scale  $\alpha$ , this is equivalently

$$X(\eta) = \frac{\eta}{\alpha} \left( I + \sqrt{I + (\eta/\alpha)^2 C C^*} \right)^{-1} C.$$

These formulas follow by minimizing each scalar function  $\alpha_i b(x) - \eta w_i x$ : its derivative equation is  $2\alpha_i x / (1 - x^2) = \eta w_i$ . The spectral trace inequalities used in [theorem 3.3](#) show that the aligned point minimizes in the whole domain. Strict convexity proves uniqueness, including tied and zero objective weights. Differentiation gives  $dx/ds = x(1 - x^2)/(1 + x^2)$  and then (4.1). In particular,  $v$  increases strictly from zero to one and

$$v(s) \leq e^s / \sqrt{2} \quad (s \leq 0), \quad 1 - v(s) \leq e^{-s} \quad (s \geq 0), \quad H(s) = s + \kappa + o(1) \quad (s \rightarrow \infty), \quad (4.3)$$

for a finite constant  $\kappa$ . The last assertion follows by integrating the first two bounds. They also show  $H(s) \leq 1/\sqrt{2} + \max(s, 0)$ .

Order channels by nondecreasing  $a_i$ , breaking ties arbitrarily, and set

$$S_0 = 0, \quad S_i = \sum_{j \leq i} \alpha_j, \quad d_i = \sqrt{S_i} - \sqrt{S_{i-1}}, \quad \Gamma_{\alpha, a}^2 = \sum_{i=1}^r \frac{d_i^2}{\alpha_i}. \quad (4.4)$$

The value of this coefficient can depend on the ordering chosen inside a tied group; any such ordering gives the following bound. Its sharpness statement allows the objective to choose strictly separated thresholds. For equal scales we write

$$\Gamma_r^2 = \sum_{i=1}^r (\sqrt{i} - \sqrt{i-1})^2 = \frac{1}{4} \log r + O(1). \quad (4.5)$$

**Theorem 4.1** (Exact worst comparison for a prescribed scale order). *For every central subarc, including the analytic-center limit,*

$$d_\Phi(x(\eta_0), x(\eta_1)) \leq L_{\text{CP}}(\eta_0, \eta_1) \leq \Gamma_{\alpha, a} d_\Phi(x(\eta_0), x(\eta_1)), \quad 0 \leq \eta_0 < \eta_1. \quad (4.6)$$

*For each fixed ordered positive scale vector, the supremum of the analytic-center-to-central-endpoint ratio, over objectives enforcing this strict activation order and over terminal parameters, is exactly  $\Gamma_{\alpha, a}$ . It is already realized as a supremum on a product of scalar intervals.*

*Proof.* For  $h_1 \geq \dots \geq h_r \geq 0$ , put  $h_{r+1} = 0$ . Decomposition into initial-coordinate indicator vectors and the triangle inequality give

$$\left( \sum_i \alpha_i h_i^2 \right)^{1/2} \leq \sum_j \sqrt{S_j} (h_j - h_{j+1}) = \sum_i d_i h_i. \quad (4.7)$$

Apply this to  $h_i = v(s - a_i)$  and integrate over the parameter interval. Weighted Cauchy–Schwarz bounds the result by  $\Gamma_{\alpha,a}(\sum_i \alpha_i (\Delta y_i)^2)^{1/2}$ . The latter norm is the exact endpoint distance by [theorem 3.1](#). In a Jordan factor, the fixed signs and order of absolute objective eigenvalues ensure a common ordered signed frame throughout the path; in a matrix factor the singular frame is likewise coordered. Thus this equality does apply to central subarcs.

For sharpness set  $c_i = d_i/\alpha_i = 1/(\sqrt{S_i} + \sqrt{S_{i-1}})$ , a strictly decreasing sequence. For  $M \rightarrow \infty$ , choose  $u_i = H^{-1}(Mc_i)$ ,  $w_i = \alpha_i e^{u_i}$ , and terminal  $s = 0$ . The endpoint distance is exactly  $M\Gamma_{\alpha,a}$ . Replacing each  $v(s + u_i)$  by  $\mathbf{1}_{s > -u_i}$  changes the integrated weighted norm on  $(-\infty, 0]$  by at most  $\sum_i \sqrt{\alpha_i} \int_{\mathbb{R}} |v - \mathbf{1}_{s > 0}| ds$ , independent of  $M$ . For large  $M$ , all  $u_i$  are positive, and the step-function length is  $\sum_i d_i u_i$ . By (4.3),  $u_i = Mc_i - \kappa + o(1)$ , so this length is  $M\Gamma_{\alpha,a}^2 + O_{\alpha}(1)$ . Taking the ratio proves the supremum. Finally,  $(\sqrt{i} - \sqrt{i-1})^2 = 1/(4i) + O(i^{-2})$  proves (4.5).  $\square$

The decreasing rearrangement norm in (4.7) is a finite weighted form of a classical Lorentz norm [18]. The elementary prefix inequality itself is not a novelty claim; the sharp central-path realization is the point of [theorem 4.1](#). Unlike the general comparison (1.2), its equal-scale constant depends on active objective rank rather than ambient dimension. The sharp limiting objectives can have large dynamic range. A separate rational-input construction is given in [section 6](#).

**Proposition 4.2** (Scale range and activation order). *For  $r \geq 2$ ,*

$$\Gamma_{\alpha,a}^2 \leq 1 + (r-1) \tanh\left(\frac{\log(S_r/S_1)}{4(r-1)}\right) \leq \min\{r, 1 + \frac{1}{4} \log(S_r/S_1)\}. \quad (4.8)$$

*Equality in the first inequality holds precisely when the cumulative ratios  $S_i/S_{i-1}$ ,  $i \geq 2$ , are equal. Among orders of a fixed multiset of scales, nondecreasing scales maximize  $\Gamma$  and nonincreasing scales minimize it.*

*Proof.* The first term of (4.4) is one; its  $i$ th term,  $i \geq 2$ , is  $\tanh(\frac{1}{4} \log(S_i/S_{i-1}))$ . Strict concavity of  $\tanh$  on  $(0, \infty)$  gives the first inequality and its equality condition. The bounds  $\tanh z \leq \min(1, z)$  give the second. For the ordering assertion, interchange adjacent scales  $a \leq b$  with preceding mass  $S > 0$ . The two logarithmic increments have common sum  $T = \log((S+a+b)/S)$ . Their first increments are  $u = \log((S+a)/S) \leq v = \log((S+b)/S)$ , and  $u+v \geq T$  because  $(S+a)(S+b) \geq S(S+a+b)$ . Thus  $|u - T/2| \leq |v - T/2|$ . Concavity makes the symmetric sum  $\tanh(u/4) + \tanh((T-u)/4)$  at least its value at  $v$ . If  $S = 0$ , the first contribution is one and the second is larger with the smaller scale first. Adjacent exchanges prove both assertions.  $\square$

For example, cumulative-geometric masses with a fixed ratio greater than one give  $\Gamma = \Theta(\sqrt{r})$ . Conversely, this order requires  $\log(S_r/\alpha_{\min}) = \Omega(r)$ . With  $\alpha_i \geq 1$  and polynomially bounded total scale, the overhead remains  $O(\sqrt{\log r})$ . For  $r = 1$ ,  $\Gamma = 1$  without a scale restriction.

## 4.2 The exact scalar comparison with the closest accurate endpoint

Define  $p(y) = b'(\rho^{-1}(y))$  for  $y \geq 0$ , using the functional inverse of  $\rho$ . Then  $p'(y) = \rho'(\rho^{-1}(y)) > 0$  and  $p$  is strictly convex on  $(0, \infty)$ .

**Theorem 4.3** (Uniform scalar dilation and the accuracy comparison). *The exact least constant satisfying  $p(cy) \geq yp'(y)$  for all  $y > 0$  is*

$$c_{\star} = \max_{y>0} \frac{p^{-1}(yp'(y))}{y}, \quad \frac{68743}{50000} < c_{\star} < \frac{69}{50}. \quad (4.9)$$

The maximum is attained at a finite positive argument. For every product under consideration and  $0 < \varepsilon < \sum_i w_i$ ,

$$L_{\text{CP}}(\varepsilon) \leq c_\star \Gamma_{\alpha,a} L_{\text{opt}}(\varepsilon). \quad (4.10)$$

The constant  $c_\star$  is sharp for this scalar inequality; no assertion is made that its product with  $\Gamma$  is the sharp arclength ratio.

*Proof.* Use  $v = \sqrt{2}x/\sqrt{1+x^2} \in (0,1)$  and write

$$\begin{aligned} Y(v) &= 2 \operatorname{artanh} v - \sqrt{2} \operatorname{artanh}(v/\sqrt{2}) = \rho(x), \\ A(v) &= \frac{\sqrt{2-v^2}}{1-v^2} = p'(Y(v)), \quad P(v) = vA(v) = p(Y(v)), \quad Y'(v) = \frac{2}{(1-v^2)(2-v^2)}. \end{aligned} \quad (4.11)$$

These identities follow by differentiation and equality at zero. Put

$$W(v) = \sqrt{1 - (1 + Y(v)^2 A(v)^2)^{-1/2}}.$$

The ratio defining  $c_\star$  is  $Y(W(v))/Y(v)$ . As  $v \downarrow 0$ ,  $Y(v) = v + O(v^3)$  and this ratio tends to one. As  $v \uparrow 1$ ,  $Y(v) = \log(1/(1-v)) + O(1)$  and  $1 - W(v) = \Theta((1-v)/Y(v))$ , so  $Y(W(v)) = Y(v) + \log Y(v) + O(1)$  and the ratio again tends to one. Strict convexity gives  $yp'(y) > p(y)$ , so the ratio exceeds one inside the interval. Continuity and these endpoint limits prove attainment. [Appendix A](#) proves both strict rational bounds.

Let  $y_i^\star = \rho(x_i^\star)$  be the exact allocation in [theorem 3.3](#), with multiplier  $\lambda > 0$ . Its equations are  $2\alpha_i y_i^\star p'(y_i^\star) = \lambda w_i$ . At central parameter  $\eta = \lambda/2$ , the central coordinates obey  $\alpha_i p(y_i^c) = \lambda w_i/2$ , hence

$$p(y_i^c) = y_i^\star p'(y_i^\star), \quad y_i^\star \leq y_i^c \leq c_\star y_i^\star.$$

This center is accurate. The first accurate center occurs no later and is no larger in any active coordinate. Its distance from zero is at most  $c_\star L_{\text{opt}}(\varepsilon)$ . Apply [theorem 4.1](#).  $\square$

The formula (4.11) gives a direct one-variable way to evaluate the constant. Numerical evaluation suggests  $c_\star \approx 1.37486420044$ ; the theorem uses only its variational definition and the rational certificates. Neither uniqueness of its maximizer nor a decimal accuracy guarantee is needed.

## 5 Objective distributions and the amount of primal movement

This section specializes to unit scales. Pool the  $r$  positive objective weights and order them as  $w_1 \geq \dots \geq w_r > 0$ . The central error and squared speed in logarithmic parameter are exactly

$$E(\eta) = \sum_i w_i (1 - q(\eta w_i)), \quad r_{\text{eff}}(\eta) = \sum_i \left( 1 - \frac{1}{\sqrt{1 + (\eta w_i)^2}} \right). \quad (5.1)$$

Thus active objective directions, rather than all spectral directions of the domain, determine primal velocity. For a matrix factor,  $r_{\text{eff}} = \operatorname{tr}[I - (I + \eta^2 C C^*)^{-1/2}]$ ; for products the traces add. Inactive directions contribute zero even if the ambient barrier parameter is large.

**Proposition 5.1** (Distribution law). *Let  $Q_j(\eta) = \sum_i \min((\eta w_i)^j, 1)$  for  $j = 1, 2$ , and set  $v_0 = \sqrt{1 - 1/\sqrt{2}}$ . Then*

$$(2 - \sqrt{2})Q_1(\eta)/\eta \leq E(\eta) \leq Q_1(\eta)/\eta, \quad (5.2)$$

$$v_0 \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta} \leq L_{\text{CP}}(\eta_0, \eta_1) \leq \int_{\eta_0}^{\eta_1} \sqrt{Q_2(\eta)} \frac{d\eta}{\eta}. \quad (5.3)$$



The pointwise bound (5.2) holds for  $\eta > 0$ . The arc bound (5.3) holds for  $0 \leq \eta_0 < \eta_1$ , including the convergent improper integral at zero.

*Proof.* The identity  $z(1 - q(z)) = 1 + z - \sqrt{1 + z^2}$  gives

$$(2 - \sqrt{2}) \min(z, 1) \leq z(1 - q(z)) \leq \min(z, 1).$$

For  $0 < z \leq 1$ , divide by  $z$  and use that  $q$  increases; for  $z \geq 1$  use monotonicity of  $1 + z - \sqrt{1 + z^2}$  and its limit one. Furthermore,

$$1 - (1 + z^2)^{-1/2} = \frac{z^2}{\sqrt{1 + z^2}(1 + \sqrt{1 + z^2})} \in [(1 - 1/\sqrt{2}) \min(z^2, 1), \min(z^2, 1)].$$

For  $z \leq 1$  bound its displayed denominator between 2 and  $2 + \sqrt{2}$ ; for  $z \geq 1$  use monotonicity. Sum and integrate.  $\square$

The integral in (5.3) is explicit on each interval between breakpoints  $1/w_i$ . If  $m$  channels have activated and  $B = \sum_{i>m} w_i^2$ , its integrand is  $\sqrt{m + B\eta^2}/\eta$ . For  $m, B > 0$  an antiderivative is

$$\sqrt{m + B\eta^2} + \sqrt{m} \log \frac{\sqrt{B}\eta}{\sqrt{m} + \sqrt{m + B\eta^2}}.$$

For  $m = 0$  it is  $\sqrt{B}\eta$ ; for  $B = 0$  it is  $\sqrt{m} \log \eta$ . Thus the distribution law can be evaluated without resolving a path in the ambient matrix space.

**Corollary 5.2** (Accuracy-dependent numerical rank). *Assume  $w_1 \leq 1$  and  $0 < \varepsilon \leq 1$ . If  $1 \leq m \leq r$  and  $\sum_{i>m} w_i \leq \varepsilon/2$ , the center at  $\eta_f = 2m/\varepsilon$  is accurate and*

$$L_{\text{CP}}(\eta_f) \leq \sqrt{(m + \varepsilon^2/4)/2} + \sqrt{2m} \log(2m/\varepsilon). \quad (5.4)$$

*Without a tail condition, the center at  $r/\varepsilon$  is accurate and has length at most  $\|w\|_2/\sqrt{2} + \sqrt{r} \log(r/\varepsilon)$ . More generally, for  $\eta_f \geq 1$  and any  $m$ , writing  $\tau_2^2 = \sum_{i>m} w_i^2$  gives*

$$L_{\text{CP}}(\eta_f) \leq \|w\|_2/\sqrt{2} + \sqrt{m + \eta_f^2 \tau_2^2/2} \log \eta_f. \quad (5.5)$$

*Proof.* The head error is at most  $m/\eta$ , and the tail error at most its weight sum. For  $\eta \leq 1$ , squared speed is at most  $\eta^2 \|w\|_2^2/2$ , giving the first length contribution. For  $1 \leq \eta \leq \eta_f$ , squared speed is at most  $m + \eta_f^2 \tau_2^2/2$ , proving (5.5). For (5.4), a sharper tail estimate uses  $\min(z^2, 1) \leq z$ : squared speed is at most  $m + \eta \sum_{i>m} w_i \leq 2m$ . Also  $\|w\|_2^2 \leq m + (\sum_{i>m} w_i)^2$ . The exact-rank statement uses  $E \leq r/\eta$  and speed at most  $\sqrt{r}$ .  $\square$

For  $0 < R < 1$ , lemma 2.1 lets us sample a central arc of length  $L$  using at most  $\lceil L/\log(1+R) \rceil$  forward- $R$  moves. These upper bounds concern primal movement. Consider a conic representation whose cone barrier restricts to the same unit-scale spectral product barrier, up to an additive constant, with the same objective and parameter  $\eta$ . Its central projection is then the path analyzed here, and primal length is measured in the restricted Hessian metric. If its full central gap is  $\nu/\eta$ , substituting  $\eta_f = \nu/\varepsilon$  in (5.5) bounds that primal length. For example,  $\tau_2 \leq \varepsilon\sqrt{m}/\nu$ ,  $w_1 \leq 1$ , and  $0 < \varepsilon \leq \min(1, \nu)$  give a primal length at most  $\sqrt{(m + m\varepsilon^2/\nu^2)/2} + \sqrt{3m/2} \log(\nu/\varepsilon)$ . This statement does not estimate the dual movement needed to attain that gap; that distinction will be treated separately.



## 5.1 Explicit lower certificates and their limits

The exact allocation is stronger than a determinant-based rank certificate, but the latter is useful for obtaining transparent orders. Let  $\bar{w}_{g,m} = (\prod_{i \leq m} w_i)^{1/m}$ .

**Proposition 5.3** (Rank-profile certificate). *For every  $0 < \varepsilon < \sum_i w_i$ ,*

$$L_{\text{opt}}(\varepsilon) \geq \max_{1 \leq m \leq r} \sqrt{m} \left[ \log \frac{m \bar{w}_{g,m}}{2\varepsilon} \right]_+. \quad (5.6)$$

*In fact the factor 2 can be omitted when this is derived directly from the exact distance formula. Neither version characterizes  $L_{\text{opt}}$  within a universal multiplicative factor.*

*Proof.* For any feasible allocation  $x_i \in [0, 1]$ , put  $e_i = w_i(1 - x_i)$ . For a subset  $I$  of size  $m$ , AM–GM gives  $\prod_{i \in I} (1 - x_i) \leq (\varepsilon/m)^m / \prod_{i \in I} w_i$ . Since  $\rho(x) \geq \log(1/(1 - x))$ , Cauchy–Schwarz gives

$$\left( \sum_i \rho(x_i)^2 \right)^{1/2} \geq \sqrt{m} \left[ \log \frac{m(\prod_{i \in I} w_i)^{1/m}}{\varepsilon} \right]_+.$$

Minimize over allocations and choose the top  $m$  weights. This proves the stronger claim and hence (5.6). For the failure of the weaker determinant certificate, fix  $0 < c < \log(2/\sqrt{e})$ ,  $\varepsilon = 1/2$ , and set

$$s_i = c/\sqrt{i}, \quad B_r = \sum_{i=1}^r i^{-1/2}, \quad a = \varepsilon/(cB_r), \quad w_i = as_i e^{s_i}.$$

The logarithmic allocation has multiplier  $1/a$ , optimizer  $s_i$ , and value  $c(\sum_i 1/i)^{1/2}$ . Meanwhile

$$\frac{m \bar{w}_{g,m}}{2\varepsilon} = \frac{m(m!)^{-1/(2m)} e^{(c/m) \sum_{i \leq m} i^{-1/2}}}{2B_r} \leq \frac{e^c \sqrt{e}}{2} \sqrt{m/r} < 1,$$

using  $m! \geq (m/e)^m$  and  $B_r \geq \sqrt{r}$ . Thus every term of (5.6) vanishes although  $L_{\text{opt}} \rightarrow \infty$ . For the stronger version, keep the same  $c < \log(2/\sqrt{e})$ . The ratio without 2 is at most  $e^c \sqrt{er}/B_r$ , uniformly in  $m \leq r$ . Since  $B_r \geq 2(\sqrt{r+1} - 1)$ , this upper bound tends to  $e^c \sqrt{e}/2 < 1$ . Thus even the stronger certificate is identically zero for all sufficiently large  $r$ .  $\square$

For completeness, the determinant origin of the factor 2 can be seen directly on a matrix ball. If  $U$  consists of  $m$  objective left singular vectors, set  $\psi_U(X) = -\log \det(I - U^* X X^* U)$ . Then

$$\nabla^2 \psi_U \preceq \nabla^2 \Phi, \quad |D\psi_U(X)[Z]| \leq \sqrt{m} \|Z\|_{\Phi, X}. \quad (5.7)$$

Indeed,  $\Phi = -\log \det \begin{pmatrix} I & X \\ X^* & I \end{pmatrix}$ ;  $\psi_U$  is the negative log determinant of its principal compression. Their difference is the negative log determinant of a Schur complement, which is convex: the Schur complement is matrix concave (the quadratic perspective  $B^* A^{-1} B$  is matrix convex), and negative log determinant is convex and order reversing. The second assertion follows from the exact parameter  $m$  of the compressed matrix-ball barrier. If  $z_i = \text{Re}(u_i^* X v_i)$ , Hadamard and AM–GM give

$$\det(I - U^* X X^* U) \leq \prod_{i \leq m} (1 - z_i^2) \leq 2^m \prod_{i \leq m} (1 - z_i) \leq \frac{(2\varepsilon/m)^m}{\prod_{i \leq m} w_i}.$$

Integrating (5.7) yields (5.6). The allocation proof avoids this compression and also works in every Jordan factor.

## 5.2 Two decay laws with matching arbitrary-path bounds

These examples require sufficiently many channels at the requested accuracy; a fixed finite rank cannot support the stated asymptotics as  $\varepsilon \downarrow 0$  indefinitely.

For geometric weights  $w_i = e^{-a(i-1)}$ , with fixed  $a > 0$ , put  $L = \log(1/\varepsilon)$ . For sufficiently large  $L$ , assume  $r \geq \lfloor L/a \rfloor$ . Then

$$L_{\text{opt}}(\varepsilon) = \Theta_a(L^{3/2}), \quad L_{\text{CP}}(\varepsilon) = \Theta_a(L^{3/2}). \quad (5.8)$$

To prove the lower bound choose  $m = \lfloor L/a \rfloor \geq 2$  in (5.6). Its logarithm is  $L - a(m-1)/2 + \log(m/2) \geq L/2$ , and  $m \geq L/(2a)$ . For the upper bound take  $M = \lceil (L + \log(2/(1 - e^{-a}))) / a \rceil$  and use corollary 5.2 with  $\min(M, r)$ . The tail is at most  $\varepsilon/2$  when  $M < r$ , and is zero otherwise;  $M = O_a(L)$ . This gives  $O_a(L^{3/2})$  length. Thus geometric decay alone need not produce a growing centrality ratio.

For polynomial weights  $w_i = i^{-b}$ , with fixed  $b > 1$ , let  $m = \lfloor (4\varepsilon)^{-1/(b-1)} \rfloor$ . If  $\varepsilon$  is sufficiently small and  $r \geq m$ , then

$$L_{\text{opt}}(\varepsilon) = \Theta_b(\varepsilon^{-1/[2(b-1)]}), \quad L_{\text{CP}}(\varepsilon) = \Theta_b(\varepsilon^{-1/[2(b-1)]}). \quad (5.9)$$

For  $\eta \geq 1$ , splitting at  $\lceil \eta^{1/b} \rceil$  and comparing tails with integrals gives  $Q_1(\eta) \leq A_b \eta^{1/b}$  and  $Q_2(\eta) \leq B_b \eta^{1/b}$ , where  $A_b = 2 + 1/(b-1)$  and  $B_b = 2 + 1/(2b-1)$ . Consequently  $\eta_f = (A_b/\varepsilon)^{b/(b-1)}$  is accurate, and (5.3) bounds its length by  $\sqrt{\zeta(2b)/2} + 2b\sqrt{B_b}(\eta_f^{1/(2b)} - 1)$ . For the lower bound,  $\bar{w}_{g,m} = (m!)^{-b/m} \geq m^{-b}$ , so the logarithm in (5.6) is at least  $\log 2$ . Since  $m = \Theta_b(\varepsilon^{-1/(b-1)})$ , the orders agree. The absence of an extra logarithm follows from the gradual activation of new channels.

## 6 A rational family and finite central-neighborhood sequences

We now separate central and arbitrary movement on the same explicit sparse linear program. In the unit-scale results preceding the final subsection,  $r \rightarrow \infty$  and  $F(x) = \sum_{i=1}^r b(x_i)$  on  $(-1, 1)^r$ . The objective is to minimize  $-w^\top x$ , equivalently to maximize the support direction  $c = w$ , so  $g_c(x) = \sum_i w_i(1 - x_i)$ . Sign changes are barrier isometries, so arbitrary prescribed objective signs give the same results.

Define the following exact rational and integer data:

$$h_j = \frac{1}{j \lceil \sqrt{j} \rceil}, \quad A_i = \left\lceil 4r \sum_{j < i} h_j \right\rceil, \quad a_i = A_i \log 2, \quad w_i = 2^{-A_i}, \quad T = a_r, \quad \varepsilon_r = r 2^{-A_r}. \quad (6.1)$$

The box has two inequalities per coordinate, with one nonzero per row and two per column. Since  $A_i = \Theta(r)$  for every  $i \geq 2$ , the ordinary rational numerator–denominator encoding of the objective has  $\Theta(r^2)$  bits, and  $\varepsilon_r$  has  $\Theta(r)$  bits. Integer square roots, rational summation and flooring construct the data in polynomial time. An exponent encoding is a different, more succinct representation.

**Theorem 6.1** (Three primal comparisons on the dyadic family). *Let  $x(s)$  denote the exact center at  $\eta = e^s$ . For all sufficiently large  $r$ , its first  $\varepsilon_r$ -accurate logarithmic parameter satisfies  $T - 1 < s_\varepsilon \leq T$ , and*

$$L_{\text{opt}}(\varepsilon_r) = \Theta(r\sqrt{\log r}), \quad d_F(0, x(s_\varepsilon)) = \Theta(r\sqrt{\log r}), \quad L_{\text{CP}}(\varepsilon_r) = \Theta(r \log r). \quad (6.2)$$

Consequently, for fixed  $0 < R < 1$ , the smallest number of arbitrary forward- $R$  moves to accuracy is  $\Theta_R(r\sqrt{\log r})$ . For the equal-scale standard barriers in rank  $r$ , the supremum of  $L_{\text{CP}}(\varepsilon)/L_{\text{opt}}(\varepsilon)$  is  $\Theta(\sqrt{\log r})$ .

*Proof.* Write  $\bar{a}_i = 4r(\log 2) \sum_{j < i} h_j$  and  $e_i = a_i - \bar{a}_i \in (-\log 2, 0]$ . The inequalities  $\frac{1}{2}j^{-3/2} \leq h_j \leq j^{-3/2}$  show  $T = \Theta(r)$  and

$$T - a_i = 4r(\log 2) \sum_{j=i}^{r-1} h_j + O(1), \quad a_{j+1} - a_j \geq 2r(\log 2)j^{-3/2} - \log 2. \quad (6.3)$$

The upper error estimate in [proposition 5.1](#) gives  $E(e^T) \leq re^{-T}$ . For a sufficiently large absolute constant  $K$ , all  $i \leq r - K\sqrt{r}$  have  $T - a_i \geq 1$ : sum the lower bound on  $h_j$  over the last  $K\sqrt{r}$  terms and use the bounded rounding error. At  $s = T - 1$ , these channels have  $e^{s-a_i} \geq 1$ , so  $E(e^{T-1}) \geq e^{-(T-1)}(2 - \sqrt{2})(r - O(\sqrt{r})) > re^{-T}$ . Strictly decreasing error proves the asserted window.

For  $u \geq -1$ , [\(4.3\)](#) gives  $H(u) = \Theta(1 + u_+)$  with universal constants. Tail summation in [\(6.3\)](#) gives  $\sum_i (T - a_i)^2 = \Theta(r^2 \log r)$ : the upper bound follows from  $T - a_i \leq Cr/\sqrt{i}$ , and the lower from  $T - a_i \geq cr/\sqrt{i}$  for  $i \leq r/16$ , after adjusting constants and discarding finitely many small  $r$ . The accurate-center window therefore proves the middle assertion of [\(6.2\)](#) and an upper bound on  $L_{\text{opt}}$ . For the lower bound on  $L_{\text{opt}}$ , every feasible allocation satisfies  $w_i(1 - x_i) \leq \varepsilon_r$ , hence  $\rho(x_i) \geq [T - a_i - \log r]_+$ . On  $i \leq \sqrt{r}$  these lower bounds are at least  $c'r/\sqrt{i}$  for large  $r$ , and their squared sum is  $\Omega(r^2 \log r)$ . Apply [theorem 3.3](#).

Let  $J = \lfloor r^{2/3} \rfloor$ . On  $[a_j, a_{j+1}]$  the first  $j$  velocities are at least  $v_0$ . These intervals for  $j \leq J$  end before  $s_\varepsilon$ , since  $T - a_{J+1} = \Theta(r^{2/3})$ . Their total length is at least  $v_0 \sum_{j \leq J} \sqrt{j}(a_{j+1} - a_j)$ . Summation by parts bounds the rounding contribution  $\sum_{j \leq J} \sqrt{j}(e_{j+1} - e_j)$  in absolute value by  $2(\log 2)\sqrt{J}$ . The unrounded sum is  $4r(\log 2) \sum_{j \leq J} \sqrt{j}h_j = \Theta(r \log r)$ . This proves the central lower bound; [theorem 4.1](#) gives its matching upper bound. [Corollary 3.4](#) gives the optimal count, and [theorem 4.3](#) gives the universal same-accuracy upper ratio. Scalar intervals embed as diagonal spectral or Jordan channels, so the lower ratio applies throughout the claimed class.  $\square$

## 6.1 Arbitrary labels, actual accuracy and a growing metric tube

For a finite sequence  $z_0, \dots, z_N$ , the following result requires the actual initial point and objective accuracy, rather than prescribed starting and ending central parameters. Reference labels may backtrack.

**Theorem 6.2** (Discrete neighborhood lower bound). *Fix  $0 < R < 1$ . For the family [\(6.1\)](#), suppose*

$$z_0 = 0, \quad g_c(z_N) \leq \varepsilon_r, \quad \|z_{k+1} - z_k\|_{F, z_k} \leq R, \quad d_F(z_k, x(s_k)) \leq \delta_r \quad (6.4)$$

*for arbitrary labels  $s_k \in \mathbb{R} \cup \{-\infty\}$ , where  $x(-\infty) = 0$ . If  $\delta_r = o(r^{2/3})$  and  $C_r = 2\delta_r - \log(1 - R)$ , then*

$$N = \Omega_R \left( \frac{r \log r}{C_r \sqrt{1 + C_r}} \right). \quad (6.5)$$

*In particular, a fixed tube has lower count  $\Omega_{R, \delta}(r \log r)$ , matching exact central-arc sampling. Relative to optimal unrestricted movement, the lower ratio diverges whenever  $\delta_r = o((\log r)^{1/3})$ .*

*Proof.* The triangle inequality and [lemma 2.1](#) bound consecutive center distances by  $C_r$ . Define

$$J = \lfloor r^{2/3} \rfloor, \quad b = a_{J+1}, \quad n(u) = \#\{i : a_i \leq u\}, \quad P(s) = \int_0^{\min(\max(s, 0), b)} \sqrt{n(u)} du.$$

For large  $r$ , the first  $J$  threshold gaps are at least  $g = \log 2$  by [\(6.3\)](#). Also  $P(b) = \Theta(r \log r)$  by the harmonic sum in the proof of [theorem 6.1](#).

Consider any pair of labels and clip both into  $[0, b]$ . Because every  $H(s - a_i)$  increases, clipping cannot increase the Euclidean distance between their coordinate vectors. Order the clipped labels as  $u \leq t$ . If  $u = t$  there is no progress change. Otherwise put  $j = n(u) \geq 1$ ,  $\Delta = t - u$ , and  $D = C_r/v_0$ . The first  $j$  coordinates have velocity at least  $v_0$  throughout  $[u, t]$ , so  $\Delta \leq D/\sqrt{j}$ . Uniform threshold spacing gives  $n(t) \leq j + 1 + \Delta/g$  (endpoint values do not affect the integral). Therefore

$$|P(s_{k+1}) - P(s_k)| \leq \Delta \sqrt{j + 1 + \Delta/g} \leq D \sqrt{2 + D/g} = O(C_r \sqrt{1 + C_r}). \quad (6.6)$$

This argument includes jumps across either clipping boundary and negative label moves.

We next force the required endpoint progress. Actual accuracy and  $w_1 = 1$  imply  $z_{N,1} \geq 1 - \varepsilon_r$ , since each coordinate's objective gap is nonnegative. Scalar metric contraction and the terminal tube give  $H(s_N) \geq \rho(1 - \varepsilon_r) - \delta_r \geq T - \log r - \delta_r$ . Since  $H(s) \leq 1/\sqrt{2} + s_+$ , this implies  $s_N \geq T - \log r - \delta_r - 1/\sqrt{2} > b$  for large  $r$ ; here  $T - b = \Theta(r^{2/3})$  is essential. For the initial progress we can bound the entire clipped step-profile arc by the smooth central arc: on  $[0, b]$  its speed  $\sqrt{n(s)}$  is at most  $\|(v(s - a_i))\|_2/v_0$ . Hence

$$P(s_0) \leq v_0^{-1} L_{\text{CP}}(e^{s_0}) \leq v_0^{-1} \Gamma_r d_F(0, x(s_0)) \leq v_0^{-1} \Gamma_r \delta_r = o(r \log r).$$

For  $s_0 \leq 0$  the first inequality is immediate because  $P(s_0) = 0$ ; for  $s_0 > 0$  it follows by integration. Thus the net progress is  $\Theta(r \log r)$ . Sum absolute increments in (6.6); no monotonicity hypothesis is used. The final ratio follows from [theorem 6.1](#) because  $C_r \sqrt{1 + C_r} = o(\sqrt{\log r})$  in the stated regime.  $\square$

The label  $-\infty$  simply includes the analytic center in the exact central sequence, even for tube radius zero. If finite labels are required, the same proof holds; an exact radius-zero tube with analytic-center initialization would then be an empty class.

**Lemma 6.3** (Newton decrement implies a metric tube). *Let  $f = F - \langle c, \cdot \rangle$  have an interior minimizer  $x^*$ , where  $F$  is standard self-concordant. Write  $\lambda_f(z) = \|\nabla f(z)\|_{F,z,*}$ . If  $\lambda_f(z) \leq \beta < 1/2$ , with  $\beta \geq 0$ , then*

$$d_F(z, x^*) \leq \delta_\beta := \log \frac{1 - \beta}{1 - 2\beta}. \quad (6.7)$$

*Proof.* Put  $h = x^* - z$  and  $t = \|h\|_{F,z}$ . Along the segment, the scalar local norm  $a(u) = \|h\|_{F,z+uh}$  satisfies  $|a'| \leq a^2$ , hence  $a(u) \geq t/(1 + ut)$ . Since  $\nabla f(x^*) = 0$ ,

$$-Df(z)[h] = \int_0^1 D^2 f(z + uh)[h, h] du \geq \int_0^1 \frac{t^2}{(1 + ut)^2} du = \frac{t^2}{1 + t}.$$

Dual Cauchy-Schwarz gives  $t \leq \lambda_f(z)/(1 - \lambda_f(z)) < 1$ . The straight-segment upper bound is  $-\log(1 - t)$ ; substituting  $\beta$  proves the claim. If  $t = 0$ , the conclusion holds directly.  $\square$

Therefore [theorem 6.2](#) applies to every forward- $R$  sequence that maintains the usual fixed decrement condition  $\|\nabla F(z_k) - e^{s_k} w\|_{F,z_k,*} \leq \beta < 1/2$  and returns an actually accurate point. It also allows  $\beta_r \uparrow 1/2$  whenever  $\delta_{\beta_r} = o((\log r)^{1/3})$  for a diverging overhead.

There is a useful precise transfer to feasible conic methods. Suppose a cone barrier  $\hat{F}$  restricts on an affine slice  $z = z_0 + Bx$  to  $F(x)$  up to an additive constant, and suppose  $q = c - A^*y$  is a feasible slack for a conic minimization problem, with  $AB = 0$  and  $B^*c = -w$ . The dual norm of the restricted covector  $B^*\ell$  in the restricted Hessian is at most the full dual norm of  $\ell$ : both

are suprema of  $\ell[h]$  over local unit vectors, and restriction reduces that set. Consequently the conventional residual condition

$$\|q + \mu \nabla \hat{F}(z)\|_{\hat{F}, z, *} \leq \beta \mu$$

implies the primal decrement bound at parameter  $1/\mu$ , since  $B^*(q + \mu \nabla \hat{F}(z))/\mu = \nabla F(x) - w/\mu$ . A bound on a full primal–dual product norm bounds its primal component and hence every restricted primal chord. Weak duality turns a feasible final duality gap at most  $\varepsilon_r$  into primal accuracy. These hypotheses transfer the lower count; they do not construct a dual-feasible version of the shorter primal route.

The box construction also yields the same lower counts on a matrix ball or Jordan interval with at least  $r$  channels, by placing the positive weights in a fixed spectral frame and setting the other objective weights to zero. Intermediate points need not stay in that frame. The contraction in [theorem 3.1](#) bounds their distance to each labeled center, and the spectral trace inequality turns final accuracy into a lower bound  $1 - \varepsilon_r$  on the largest singular value or Jordan eigenvalue. These are exactly the two facts about arbitrary iterates used in the proof above.

## 6.2 Unequal scales can produce a larger discrete separation

The scale frontier also has a finite-sequence interpretation, with an explicit parameter-crossing assumption distinct from [\(6.4\)](#). In this subsection the barrier is  $F_\alpha(x) = \sum_{i=1}^r \alpha_i b(x_i)$  on  $(-1, 1)^r$ ; all local norms, distances, lengths, and labeled centers refer to this weighted barrier. Fix  $B \geq 4$  and an integer  $r$ . For large  $U$ , put

$$u_i = UB^{r-i}, \quad H_0 = H(u_1), \quad \alpha_i = \left( \frac{H_0}{H(u_i)} \right)^2 \geq 1, \quad w_i = \alpha_i e^{u_i}, \quad i = 1, \dots, r.$$

At terminal label zero all weighted radial coordinates equal  $H_0$ ; thus the endpoint distance is  $H_0\sqrt{r}$ . The central length is  $\Theta(H_0r)$ , uniformly for large  $U$ : for fixed  $M > 0$  the disjoint cores  $I_i = [-u_i + M, -u_{i+1} - M]$ ,  $i < r$ , and  $I_r = [-u_r + M, 0]$  each have gain  $\Theta(H_0)$  in their  $i$ th weighted radial coordinate. Indeed,  $H(t) = t + O(1)$  for positive  $t$  makes this gain at least a fixed fraction of  $H_0$ ; the upper bound follows from [theorem 4.1](#) and  $\Gamma \leq \sqrt{r}$ .

For fixed  $0 < R < 1$  and  $\delta \geq 0$ , consider forward- $R$  iterates starting at  $z_0 = 0$  and within distance  $\delta$  of their labeled centers, with  $s_0 \leq -u_1 + M$  and  $s_N \geq 0$ ; the initial label may be  $-\infty$ . They need  $\Omega_{R,\delta}(H_0r)$  moves once  $U$  is large enough. To see this, sum the clipped  $i$ th coordinate gain across each core as a scalar potential. Every full core has gain exceeding  $C = 2\delta - \log(1 - R)$ . A permitted center-to-center jump can meet at most two cores in positive length: meeting three would cross the middle one fully, contradicting its distance bound. The at most two clipped coordinate gains sum to at most  $\sqrt{2}C$  by Cauchy–Schwarz and exact coordinate distance. Net progress is  $\Theta(H_0r)$ , even with backtracking. To compare the same endpoints, specialize to  $z_N = x(0)$  and  $s_N = 0$ . The lower bound still applies, while sampling a shortest route from 0 to  $x(0)$  takes  $O_R(H_0\sqrt{r})$  moves. This construction uses large scale and objective dynamic ranges; [\(4.8\)](#) explains why such ranges are needed. No ordinary-input efficiency claim is made for this construction.

## 7 Dependence on the barrier

The preceding logarithmic-rank comparison is not a consequence of self-concordance alone. Its decisive ingredient is ordered metric velocity. We first identify a scalar normalization that guarantees this property, and then show that the same order of centrality cost persists for explicit coupled barriers with the smallest possible cube parameter.

## 7.1 Every fixed normalized scalar barrier has the same sharp factor

Let  $f$  be a  $C^3$  standard self-concordant barrier on an open interval  $I = (\ell, b_f)$ , with  $f'' > 0$  and an analytic center  $x_f \in I$ . Assume that  $f'$  maps  $(x_f, b_f)$  onto  $(0, \infty)$  and that

$$(f'(x))^2 \leq f''(x) \quad (x_f < x < b_f). \quad (7.1)$$

The normalization is only required on this positive branch. Define

$$\rho_f(x) = \int_{x_f}^x \sqrt{f''(u)} du, \quad h_f(t) = \rho_f((f')^{-1}(e^t)), \quad v_f = h'_f.$$

The coordinate change is an isometry for the scalar Hessian metric; therefore the product  $F_f(x) = \sum_{i=1}^r f(x_i)$  has Euclidean distance in the coordinates  $\rho_f(x_i)$ . Coordinates below  $x_f$  can be moved to  $x_f$  while decreasing distance from the center and improving every positive linear objective. Zero objective coordinates remain at  $x_f$ ; below,  $r$  denotes the number of positive weights.

**Theorem 7.1** (Normalized scalar comparison). *Under these hypotheses, every positive-objective central subarc satisfies*

$$L_{\text{CP}}(\eta_0, \eta_1) \leq \Gamma_r d_{F_f}(x(\eta_0), x(\eta_1)).$$

For each fixed  $f$  and  $r$ , the supremum of the ratio from the analytic center, over positive objectives and terminal parameters, equals  $\Gamma_r$ . Moreover  $b_f < \infty$ , and for  $0 < \varepsilon < (b_f - x_f) \sum_i w_i$ ,

$$L_{\text{CP}, F_f}(\varepsilon) \leq C_{\text{sc}} \Gamma_r L_{\text{opt}, F_f}(\varepsilon), \quad C_{\text{sc}} < 2. \quad (7.2)$$

Here  $L_{\text{opt}, F_f}$  is distance from  $(x_f, \dots, x_f)$  to  $\sum_i w_i(b_f - x_i) \leq \varepsilon$ , and the dimension-independent constant is the explicit attained maximum in [proposition B.1](#). Numerically,  $C_{\text{sc}} \approx 1.831856423$ .

*Proof.* At  $x = (f')^{-1}(e^t)$ , differentiation gives

$$v_f = \frac{f'}{\sqrt{f''}}, \quad v'_f = v_f \left( 1 - \frac{v_f f'''}{2(f'')^{3/2}} \right) \geq v_f(1 - v_f) \geq 0. \quad (7.3)$$

Thus  $v_f$  is nondecreasing and  $0 < v_f \leq 1$ . Its limit at  $-\infty$  is zero, by continuity at  $x_f$ . Its positive limit at  $+\infty$  must be one: a smaller limit would make the derivative eventually bounded below by a positive number. The negative tail is integrable because  $\int_{-\infty}^t v_f = h_f(t) < \infty$ . Once  $v_f \geq 1/2$ , the differential inequality implies  $(1 - v_f)' \leq -(1 - v_f)/2$ . Consequently

$$v_f - \mathbf{1}_{(0, \infty)} \in L^1(\mathbb{R}), \quad h_f(t) = t + \kappa_f + o(1) \quad (t \rightarrow \infty). \quad (7.4)$$

Central metric coordinates are  $h_f(s + \log w_i)$ . Their velocities have the order of the weights, so the prefix proof of [theorem 4.1](#) applies. For sharpness, set  $h_f(\log w_i) = M(\sqrt{i} - \sqrt{i-1})$  and terminate at  $s = 0$ . The integrable step replacement in that proof has error  $O_{f,r}(1)$ ; distance is  $M\Gamma_r$  and length is  $M\Gamma_r^2 + O_{f,r}(1)$ . This also proves that the supremum is for each fixed profile, rather than for a profile varying with dimension.

Writing  $x(t) = (f')^{-1}(e^t)$  gives

$$x'(t) = e^{-t} v_f(t)^2, \quad b_f - x(t) = \int_t^\infty e^{-u} v_f(u)^2 du. \quad (7.5)$$

The first formula makes the right endpoint finite; surjectivity of  $f'$  identifies the limit with  $b_f$ . The second follows by integration. It remains to prove the accuracy estimate. Put  $p_f(y) = f'(\rho_f^{-1}(y))$  on  $y \geq 0$ . Then

$$p_f(0) = 0, \quad p'_f > 0, \quad |p''_f| \leq p'_f, \quad p_f \leq p'_f. \quad (7.6)$$



A closest accurate metric vector  $y^* \geq 0$  exists: intersect the closed feasible set with a ball containing one accurate vector, and use compactness and the finite-coordinate inverse  $\rho_f^{-1}$ . The constraint is active, since decreasing a positive coordinate otherwise reduces the norm. Every coordinate is positive. Indeed, increasing a zero coordinate by  $t$  improves the gap to first order while increasing squared norm only to second order; compensating by decreasing a positive coordinate preserves the gap and reduces squared norm to first order. The constraint gradient is nonzero. Necessary Lagrange multiplier conditions at this global minimizer therefore give

$$y_i^* p'_f(y_i^*) = \lambda w_i, \quad \lambda > 0. \quad (7.7)$$

No convexity or uniqueness of this allocation is asserted for a general  $f$ . With  $k = \log 2$ , [proposition B.1](#) proves

$$p_f(y) \leq y p'_f(y)/k \leq p_f(C_{\text{sc}} y).$$

The center at  $\eta = \lambda/k$  thus has metric coordinates between  $y_i^*$  and  $C_{\text{sc}} y_i^*$ . It is accurate, and the first accurate center is coordinatewise no farther. Apply the endpoint bound.  $\square$

More generally, the exact local test for monotonicity is  $2(f'')^2 - f' f''' \geq 0$  on the positive branch, as follows from (7.3). Monotonicity alone proves the  $\Gamma_r$  upper bound; the normalization additionally supplies the unit limiting speed, integrable tails, and uniform accuracy comparison.

**Proposition 7.2** (The full self-concordant scalar class). *Over smooth self-concordant interval barriers with an analytic center and full positive gradient range, the supremum of the analytic-center-to-central-endpoint ratio in  $r$  identical product coordinates is  $\sqrt{r}$ . In the family witnessing sharpness, the scalar barrier parameter is  $\nu_A = \Theta(A^2)$  and  $A$  tends to infinity. Thus this assertion does not concern one fixed normalized barrier or a dimension-independent parameter budget.*

*Proof.* Every central metric coordinate is increasing, whether or not its velocity is monotone. Hence  $L \leq \int \|\dot{y}\|_1 = \|y_{\text{end}}\|_1 \leq \sqrt{r} \|y_{\text{end}}\|_2$ . [Appendix B.1](#) gives a globally smooth barrier family and disjoint coordinate-velocity windows approaching equality.  $\square$

## 7.2 The same dyadic separation for every normalized profile

**Corollary 7.3** (Fixed-profile finite sequences). *Fix  $f$  as in [theorem 7.1](#). Use exactly the weights and tolerance (6.1), with objective gap  $\sum_i w_i(b_f - x_i)$  and initial point  $(x_f, \dots, x_f)$ . Then*

$$L_{\text{opt}, F_f}(\varepsilon_r) = \Theta_f(r \sqrt{\log r}), \quad L_{\text{CP}, F_f}(\varepsilon_r) = \Theta_f(r \log r).$$

*For  $0 < R < 1$ , arbitrary-label sequences obeying (6.4) in this metric, with the stated new initial point and objective gap, satisfy (6.5) whenever  $\delta_r = o(r^{2/3})$ . The constants may depend on  $f$  and  $R$ . In particular, every fixed metric tube, and every fixed Newton-decrement neighborhood with radius  $0 \leq \beta < 1/2$ , has a  $\Theta(\sqrt{\log r})$  overhead in bounded movement.*

*Proof.* Choose  $\tau_f$  so that  $v_f(t) \geq 1/2$  for  $t \geq \tau_f$ . By (7.5), each central coordinate has gap at most  $e^{-t}$ , and at least  $e^{-t}/4$  when  $t \geq \tau_f$ . Thus the center at  $T$  is accurate and the one at  $T - \log 8$  is not, for large  $r$ : only  $O_f(\sqrt{r})$  terminal coordinates fail the lower-tail condition, by (6.3). From (7.4),  $|h_f(t) - t_+| \leq B_f$  for all  $t$ . Therefore the accurate-center distance is  $\Theta_f(r \sqrt{\log r})$ . Every accurate endpoint has  $b_f - x_i \leq \varepsilon_r/w_i$ , which implies

$$\rho_f(x_i) \geq T - a_i - \log r - O_f(1)$$



whenever the right side is large. To see this implication directly, write  $x_i = x(t)$  near  $b_f$  and combine  $b_f - x(t) \geq e^{-t}/4$  with  $h_f(t) = t + O_f(1)$ . The first  $\lfloor \sqrt{r} \rfloor$  coordinates give the matching lower bound on target distance, as in [theorem 6.1](#).

Use the shifted thresholds  $a_i + \tau_f$  and cutoff  $a_{J+1} + \tau_f$ , where  $J = \lfloor r^{2/3} \rfloor$ . The first  $j$  velocities exceed  $1/2$  after threshold  $j$ . The gap spacing and harmonic sum in (6.3) are unchanged by this common shift. They give central length  $\Omega_f(r \log r)$  before the first accurate center; the prefix bound gives the matching upper bound. For discrete sequences define the clipped potential  $P_f$  by replacing  $0, b$  in the proof of [theorem 6.2](#) with  $\tau_f, a_{J+1} + \tau_f$ . The exact product metric and velocity lower bound give each progress increment  $O_f(C_r \sqrt{1 + C_r})$ , by the same calculation as (6.6), with  $v_0$  replaced by  $1/2$ . The start tube gives  $P_f(s_0) \leq 2\Gamma_r \delta_r$  by the prefix bound. Actual accuracy in coordinate one forces  $s_N \geq T - \log r - \delta_r - O_f(1) > a_{J+1} + \tau_f$ . Hence net progress is  $\Theta(r \log r)$  even with backtracking labels. Finally the straight segment in  $\rho_f$  coordinates to the center at  $T$  has length  $O_f(r \sqrt{\log r})$ ; chord sampling gives the unrestricted upper count. [Lemma 6.3](#) applies to the product barrier and proves the decrement assertion.  $\square$

The objective weights and tolerance in this corollary are dyadic with the same encoding lengths as before. An arbitrary fixed  $f$  need not have an efficient finite arithmetic oracle; this is a statement about geometric movement.

### 7.3 Universal and entropic cube barriers

The universal barrier of Nesterov and Nemirovski and the entropic barrier of Bubeck and Eldan are canonical constructions for general convex bodies. The exact dimension bounds were proved by Lee and Yue and by Chewi, respectively [\[5, 7, 15\]](#). Their cube specializations illustrate the scope of [theorem 7.1](#).

**Proposition 7.4** (Canonical barriers on a cube). *For  $K = (-1, 1)^r$ , the universal barrier is  $U(x) = \sum_i b(x_i)$  up to an additive constant. The entropic barrier is  $E(x) = \sum_i e(x_i)$ , where*

$$e = a^*, \quad a(\theta) = \log \frac{2 \sinh \theta}{\theta}, \quad a(0) = \log 2.$$

*Both have exact parameter  $r$ . For each, the worst endpoint ratio is exactly  $\Gamma_r$ , the worst same-accuracy ratio has order  $\sqrt{\log r}$ , and [corollary 7.3](#) applies.*

*Proof.* In orthant  $\sigma \in \{-1, 1\}^r$ , the polar of  $K - x$  is the simplex  $\sum_i (1 - \sigma_i x_i) |z_i| \leq 1$ , of volume  $[r! \prod_i (1 - \sigma_i x_i)]^{-1}$ . Summing gives

$$\text{vol}((K - x)^\circ) = \frac{1}{r!} \prod_i \left( \frac{1}{1 - x_i} + \frac{1}{1 + x_i} \right) = \frac{2^r}{r!} \prod_i (1 - x_i^2)^{-1}.$$

Its logarithm is the claimed universal barrier. Fubini's theorem gives  $\log \int_K e^{\langle \theta, z \rangle} dz = \sum_i a(\theta_i)$ ; separating the supremum in Fenchel conjugacy gives the entropic formula. These factorization identities are elementary specializations, not new barrier constructions; product consistency is also explicit in Chewi's tensorization argument [\[7, Section 4\]](#). The explicit cube log-partition and its coordinatewise gradient also appear in recent work on linear bandits [\[16, Section 6.1\]](#).

For the entropic summand,  $a'' > 0$  and

$$x = a'(\theta) = \coth \theta - 1/\theta, \quad a''(\theta) = 1/\theta^2 - 1/\sinh^2 \theta,$$

with their continuous values at zero. The map  $a'$  increases from  $-1$  to  $1$ , and  $e'(x) = \theta$ . Chewi's theorem gives standard self-concordance with parameter at most one in dimension one. Directly, the squared local gradient norm is

$$\frac{e'(x)^2}{e''(x)} = \theta^2 a''(\theta) = 1 - (\theta / \sinh \theta)^2 \longrightarrow 1 \quad (\theta \rightarrow \infty).$$

Thus the scalar parameter is exactly one and its positive branch satisfies (7.1); its center is zero. The product parameter is exactly  $r$  by taking all coordinates to the positive endpoint. The same facts for  $b$  were proved in proposition 3.2. Apply the normalized scalar theorem and its dyadic corollary.  $\square$

Alternative interval barriers and their Hessian geometry have substantial prior literature. Nesterov and Todd [21, Section 6.2] use  $-\log \cos$  on a rescaled interval, while Papa Quiroz and Oliveira [23] construct a different cube barrier with parameter  $3r/2$  and study geometric algorithms. Neither scalar flattening nor changing the cube barrier is new here. The conclusion above concerns the sharp central-arc comparison and the stated finite-sequence separation for the two canonical barriers.

## 7.4 Transfer to trace-separable spectral metrics

The scalar argument extends to the spectral setting without an assumption that commuting paths are globally shortest. That assertion follows from the Hessian itself. If  $F(x) = \text{tr } f(x) = \sum_i f(\lambda_i(x))$  on a Jordan spectral interval, its Peirce Hessian is

$$D^2 F(x)[u, u] = \sum_i f''(\lambda_i) u_i^2 + \sum_{i < j} \frac{f'(\lambda_i) - f'(\lambda_j)}{\lambda_i - \lambda_j} \|u_{ij}\|^2. \quad (7.8)$$

The continuous divided difference is used at coincidences, with the Peirce normalization in (3.13). One can obtain (7.8) directly: polynomial Jordan calculus gives  $D \text{tr } f(x)[u] = \langle f'(x), u \rangle$  and makes  $Df'(x)$  act on  $V_{ij}$  by the displayed divided difference. Approximate  $f''$  uniformly by polynomials on a compact interval containing the spectrum and integrate twice. The resulting polynomial functions and their first two derivatives converge uniformly; their divided differences do also, by their integral representation. This proves the formula for  $C^2$  functions. It is the trace-separable case of the classical spectral differential calculus [4, 17].

All off-frame terms are nonnegative. The almost-everywhere spectral derivative argument in theorem 3.1 consequently gives

$$\|\dot{x}\|_{F,x}^2 \geq \sum_i f''(\lambda_i) \dot{\lambda}_i^2 = \left\| \frac{d}{dt} \rho_f(\lambda(x)) \right\|^2.$$

Thus center-to-point distance, and distance between coordered endpoints in one frame, is Euclidean in the transformed spectra. Positive spectral support objectives have central equations  $f'(\lambda_i) = \eta w_i$  and a common objective frame, so theorem 7.1 applies, with active spectral rank. For the accuracy bound, the Jordan trace inequality aligns an endpoint with the ordered objective; replacing coordinates below  $x_f$  by  $x_f$  improves the objective and distance. The scalar minimizer and necessary conditions (7.7) then give the same bound.

For rectangular matrices, assume  $f$  is even on  $(-b_f, b_f)$ ,  $x_f = 0$ , and  $f'' > 0$ . Apply the Hermitian dilation argument with  $g = f/2$  to  $F(X) = \sum_i f(\sigma_i(X))$ ; the two copies of each singular value recover exactly  $f''$ . The same conclusions hold. These are assertions about Hessian metrics.

Standard self-concordance of an arbitrary spectral lift is not inferred from scalar self-concordance. Bounded-move corollaries for a lift require that property separately; [proposition 3.2](#) supplies it for the standard spectral barriers treated earlier.

## 8 Coupled cube barriers with optimal parameter

We now retain a growing centrality cost while changing the metric by genuinely coupled curvature. Throughout this section  $K = (-1, 1)^r$  and  $U(x) = \sum_i b(x_i)$ . The parameter  $r$  is the smallest possible parameter of any standard self-concordant barrier on this cube; this classical lower bound follows from its simple vertices [\[20, Proposition 2.3.6\]](#).

### 8.1 A lower bound from one regular facet collar

**Theorem 8.1** (Persistence under bounded curvature near a facet). *Let  $F = U + G$  be a standard self-concordant barrier, with  $G \in C^2(K)$  convex. Suppose that, after a signed coordinate permutation, some  $\delta \in (0, 1)$  and finite  $M, C$  satisfy*

$$\|\nabla G(x)\|_\infty \leq M, \quad 0 \leq G''(x) \leq CI \quad \text{for every } x \in K \text{ with } x_1 \geq 1 - \delta. \quad (8.1)$$

Let  $x_*$  be the analytic center of  $F$ . Then

$$\sup_{w>0, \eta>0} \frac{L_{CP,F}(0, \eta)}{d_F(x_*, x_w(\eta))} \geq \Gamma_r, \quad \sup_{w>0, 0<\varepsilon<g_w(x_*)} \frac{L_{CP,F}(\varepsilon)}{L_{\text{opt},F}(\varepsilon)} \geq \Gamma_r. \quad (8.2)$$

*Distances to accuracy in the second formula start at  $x_*$ . The objectives are positive in the transformed coordinate system. No matching upper bound for all  $G$  satisfying (8.1) is asserted.*

*Proof.* The bounded cube and barrier blow-up give an interior minimizer, unique because  $F'' \succeq U'' \succ 0$ . Let  $d_i = \sqrt{i} - \sqrt{i-1}$ , choose  $w_i = e^{Md_i}$  with  $M \rightarrow \infty$ , and write  $x_M(s)$  for the center at  $\eta = e^s$ , terminating at  $s = 0$ . In this proof only, denote the gradient bound in [\(8.1\)](#) by  $M_G$  to distinguish it from  $M$ . Set  $z_i = e^{s+Md_i}$ . For each fixed  $x_{-1}$ , convexity implies

$$x_1 \leq 1 - \delta \implies \partial_1 G(x_1, x_{-1}) \leq \partial_1 G(1 - \delta, x_{-1}) \leq M_G.$$

Choose  $Z > M_G + b'(1 - \delta)$ . The equation  $b'(x_1) + \partial_1 G = z_1$  shows that  $z_1 \geq Z$  forces entry into the collar. As  $z_1$  increases, the path stays there.

In the collar, put  $D = U''(x_M)$  and  $H = F''(x_M)$ . Differentiated stationarity gives  $H\dot{x}_M = z$  and speed squared  $z^T H^{-1} z$ . For any coordinate subset  $A$ , the variational formula for this inverse quadratic form, restricted to vectors supported on  $A$ , gives

$$z^T H^{-1} z \geq z_A^T H_{AA}^{-1} z_A \geq \sum_{i \in A} \frac{z_i^2}{b''(x_{M,i}) + C}. \quad (8.3)$$

Also  $b'(x_{M,i}) = z_i - \partial_i G$  with  $|\partial_i G| \leq M_G$ . Because  $(b')^2/b'' = 2x^2/(1+x^2) \rightarrow 1$  as  $x \uparrow 1$ , increasing  $Z$  makes every summand in [\(8.3\)](#) at least  $1 - \zeta$  whenever  $z_i \geq Z$ , for any fixed  $\zeta > 0$ . The activation times  $\tau_i = \log Z - Md_i$  are strictly ordered, and all lie below zero for large  $M$ . Integrating over their successive prefix intervals gives

$$L_{CP,F}(0, 1) \geq \sqrt{1 - \zeta} [M\Gamma_r^2 - \sqrt{r} \log Z]. \quad (8.4)$$

At termination, stationarity and the bounded gradient imply  $y_i := \rho(x_{M,i}(0)) = Md_i + O_{r,M_G}(1)$ , by (4.3); hence  $\|y\| = M\Gamma_r + O(1)$ .

Fix an interior point  $p$  with  $p_1 = 1 - \delta/2$  and a fixed compact interior curve from  $x_*$  to  $p$ . Its  $F$ -length is finite. The straight segment from  $\rho(p)$  to  $y$  in product metric coordinates has monotone physical coordinates. For large  $M$  its first coordinate stays above  $1 - \delta$ , so it lies in the collar. Its  $U$ -length is  $M\Gamma_r + O(1)$  and its Euclidean total variation is at most  $2r$ . Its extra  $G$ -length is therefore at most  $2r\sqrt{C}$ . Using  $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$  gives

$$d_F(x_*, x_M(0)) \leq M\Gamma_r + O_{F,r,M_G,C}(1). \quad (8.5)$$

Divide (8.4) by this bound and let first  $M \rightarrow \infty$ , then  $\zeta \downarrow 0$ . This proves the first assertion, with analytic-center initialization interpreted as the limit  $s \rightarrow -\infty$ . Near that limit the path has finite length because  $x(\eta) = x_* + \eta F''(x_*)^{-1}w + o(\eta)$ .

For each such objective and terminal center, choose  $\varepsilon = g_w(x_M(0)) > 0$ . The identity

$$\frac{d}{d\eta} w^T x_w(\eta) = w^T F''(x_w)^{-1}w > 0$$

shows this is the first accurate center and  $\varepsilon < g_w(x_*)$ . Since  $L_{\text{opt},F}(\varepsilon) \leq d_F(x_*, x_M(0))$ , the same construction proves the second assertion.  $\square$

The full collar in (8.1) matters: it permits the entry argument while the other coordinates remain far from their target facets. Bounded derivatives only in a small neighborhood of one vertex do not supply that implication. Within the stated convex perturbation class, an improvement to a uniformly bounded worst ratio would therefore require derivative singularities on every full signed facet collar. This is a necessary condition, not an existence claim.

## 8.2 Two explicit coupled barriers with the cube parameter

**Theorem 8.2** (Dense and radial coupling at the cube parameter). *For  $r \geq 2$ , each of the barriers*

$$F_d(x) = U(x) + \frac{1}{8r}(\mathbf{1}^T x)^2, \quad (8.6)$$

$$F_{\lambda,c}(x) = U(x) - \lambda \log(c - \|x\|^2), \quad \lambda \geq 1, \quad c - r \geq 4\lambda, \quad (8.7)$$

*is standard self-concordant with exact parameter  $r$  and has nonzero mixed Hessian entries on an open subset of the cube. The dense barrier is invariant under permutations and simultaneous sign reversal. The radial family is invariant under every signed permutation. Both endpoint and same-accuracy suprema in (8.2) are at least  $\Gamma_r$  for each fixed choice of their parameters.*

*Proof.* Write  $p = \nabla U$  and  $D = U''$ . For the dense example put  $\beta = 1/(4r)$ ,  $t = \mathbf{1}^T x$ ,  $A = \mathbf{1}^T D^{-1} \mathbf{1}$ ,  $B = \mathbf{1}^T D^{-1} p$ , and  $Q = p^T D^{-1} p$ . Sherman–Morrison gives

$$\|\nabla F_d\|_{F_d, x,*}^2 = Q + \frac{\beta(2tB - B^2) + \beta^2 t^2 A}{1 + \beta A}.$$

Let  $m = r - Q = \sum_i (1 - x_i^2)/(1 + x_i^2) > 0$ . The explicit interval derivatives give  $|B| \leq m$ ,  $A \leq m/2$ , and  $|t| < r$ . Dropping  $-\beta B^2$  bounds the correction by  $(2\beta r + \beta^2 r^2/2)m = 17m/32 < m$ . Thus the parameter is at most  $r$ . The quadratic has zero third derivative and positive semidefinite Hessian, so it preserves standard self-concordance;  $U$  supplies boundary blow-up.

For the radial case put  $S = c - \|x\|^2$  and  $a = 2\lambda/S$ . Then  $0 < a < 1/2$  and

$$\nabla F_{\lambda,c} = p + ax, \quad F''_{\lambda,c} = D + aI + (a^2/\lambda)xx^T. \quad (8.8)$$

The ball term is an affine rescaling of the rank-one spectral-ball barrier in [proposition 3.2](#); multiplying by  $\lambda \geq 1$ , restricting to the cube, and summing preserve standard self-concordance. Inverse Loewner order gives

$$\|\nabla F_{\lambda,c}\|_{F_{\lambda,c},x,*}^2 \leq \sum_i \frac{(p_i + ax_i)^2}{D_i + a} \leq r.$$

For the last inequality set  $y = x_i^2$ . After multiplying by  $(1-y)^2$ , the difference between denominator and numerator in its  $i$ th summand is

$$(1-y)[2 + a(1-5y) - a^2y(1-y)].$$

The bracket decreases on  $[0, 1]$ , since its derivative is  $a(2ay - a - 5) < 0$ , and its value at one is  $2 - 4a \geq 0$ .

For both examples, along  $x = t\mathbf{1}$  as  $t \uparrow 1$  the added gradient and Hessian stay bounded, while the interval gradient and Hessian diverge. The directional squared gradient ratio along  $\mathbf{1}$  tends to  $r$ ; explicitly it is  $r(p_0 + at)^2/[D_0 + a + (a^2/\lambda)rt^2]$  for the radial example. Thus both exact parameters equal  $r$ . The stated symmetries follow from the formulas. The mixed Hessians are  $\beta$  in the dense case and  $(a^2/\lambda)x_i x_j$  in the radial case. Finally the dense coupling obeys  $\|\nabla G\|_\infty \leq 1/4$ ,  $G'' \preceq I/4$ . The radial coupling obeys

$$\|\nabla G\|_\infty < 1/2, \quad 0 \preceq G'' \preceq (1/2 + r/(4\lambda))I.$$

These global bounds imply (8.1); apply [theorem 8.1](#).  $\square$

The generic sum certificate for (8.7) is  $r + \lambda$ , whereas its exact parameter is  $r$ . At fixed  $r$ , choosing  $c = r + 4\lambda$  allows an arbitrarily large additive overestimate. The parameter  $c$  must vary here; fixed  $r, c$  only permits  $\lambda \leq (c - r)/4$  under our sufficient condition. Quadratic regularization that preserves the box parameter is prior work: Castro and Cuesta [6, Section 4.1] prove this for appropriately bounded diagonal regularizations. The present dense rank-one example and the radial family are used to establish the simultaneous geometric comparisons, not to claim that parameter-preserving regularization itself is new.

### 8.3 A uniform matching upper bound for the radial family

**Theorem 8.3** (Radial centrality comparison). *For every barrier (8.7) and every nonzero linear objective with  $k$  nonzero coordinates, every central subarc obeys*

$$L_{\text{CP},F}(\eta_0, \eta_1) \leq B_r \Gamma_k d_F(x(\eta_0), x(\eta_1)), \quad B_r = \sqrt{\frac{11}{8}} \frac{25}{14}. \quad (8.9)$$

Starting at zero, for  $0 < \varepsilon < \sum_i |w_i|$ ,

$$L_{\text{CP},F}(\varepsilon) \leq \frac{5}{4} B_r c_\star \Gamma_k L_{\text{opt},F}(\varepsilon). \quad (8.10)$$

The constants are independent of  $r, \lambda, c$ ; their numerical prefactors are not claimed sharp. For positive objectives, together with [theorem 8.1](#), these inequalities give matching  $\Theta(\sqrt{\log r})$  worst endpoint and same-accuracy ratios for every member of this explicitly coupled optimal-parameter family.

*Proof.* Let  $t = \sum_i (1 - x_i^2)$ , so  $S \geq 4\lambda + t$ . The interval Hessian satisfies  $D \succeq 2I$  and  $x^T D^{-1} x \leq t/2$ . The two relative contributions in (8.8) satisfy

$$aI \preceq \frac{1}{4}D, \quad \frac{a^2}{\lambda}xx^T \preceq \frac{2\lambda t}{(4\lambda + t)^2}D \preceq \frac{1}{8}D.$$

The last inequality follows from  $(t - 4\lambda)^2 \geq 0$ . Consequently, throughout the whole cube,

$$U'' \preceq F'' \preceq \frac{11}{8}U''. \quad (8.11)$$

In particular  $d_F \geq d_U$ , even for paths leaving the active coordinate subspace.

By signed-permutation invariance take the nonzero weights positive and decreasing. The stationary equations at  $\eta = e^s$  are

$$z_i = e^s w_i = x_i(b_i + a), \quad b_i = 2/(1 - x_i^2). \quad (8.12)$$

Zero-weight coordinates are zero. Every other coordinate is positive, and their order is that of  $w_i$ , since  $x \mapsto x(b(x) + a)$  is strictly increasing for fixed  $a$ . Differentiating with respect to  $s$  and solving for  $a'$  gives

$$(D_i + a)x'_i + x_i a' = z_i, \quad a' = \frac{(a^2/\lambda)K}{1 + (a^2/\lambda)L}, \quad K = \sum_i \frac{x_i^2(b_i + a)}{D_i + a}, \quad L = \sum_i \frac{x_i^2}{D_i + a}. \quad (8.13)$$

For  $y = x_i^2$ , the inequality  $y(b_i + a)/(D_i + a) \leq 1 - y$  reduces, after multiplication by  $1 - y$ , to  $2 + a(1 - 2y)(1 - y) > 0$ . Thus  $K \leq t$  and

$$0 \leq a' \leq \frac{4\lambda t}{(4\lambda + t)^2} \leq \frac{1}{4}. \quad (8.14)$$

Put  $q_i = b'(x_i) = b_i x_i$  on active coordinates and define

$$\theta_i = (\log q_i)' = \frac{D_i(b_i + a - a')}{(D_i + a)b_i}, \quad v_i = \frac{q_i}{\sqrt{D_i}} = \frac{\sqrt{2}x_i}{\sqrt{1 + x_i^2}}.$$

Since  $D_i, b_i \geq 2$ ,  $a \leq 1/2$ , and  $a' \leq 1/4$ ,

$$\frac{7}{10} = \frac{4}{5} \frac{7}{8} \leq \theta_i \leq \frac{5}{4}, \quad y'_i := (\rho(x_i))' = \theta_i v_i. \quad (8.15)$$

In particular all coordinates increase. The  $v_i$  are nonnegative and decreasing in index, since the  $x_i$  are ordered. Let  $d_i = \sqrt{i} - \sqrt{i-1}$ . Combining the metric comparison, the bounds on  $\theta_i$ , and the prefix inequality yields

$$\begin{aligned} L_{\text{CP},F} &\leq \sqrt{11/8} \int \|y'\|_2 ds \leq \sqrt{11/8} \frac{5}{4} \int \|v\|_2 ds \\ &\leq \sqrt{11/8} \frac{5}{4} \sum_{i=1}^k d_i \int v_i ds \leq B_r \sum_{i=1}^k d_i \Delta y_i \leq B_r \Gamma_k \|\Delta y\|_2. \end{aligned}$$

The last norm equals the full-cube  $U$ -distance of the two centers and is at most their  $F$ -distance. This proves (8.9), including the center limit.

For accuracy, compare with the standard  $U$  center for the same weights. Equation (8.12) gives

$$\frac{4}{5}z_i \leq q_i \leq z_i. \quad (8.16)$$

Let  $\eta_U$  be the first accurate  $U$  parameter. The radial center at  $\bar{\eta} = 5\eta_U/4$  is coordinatewise at least this  $U$  center, so is accurate. By coordinate monotonicity the first accurate radial center occurs no later. For the standard profile  $H$  in (4.1),  $v = H'$  has  $v' \leq v$  because  $b''' \geq 0$  on  $(0, 1)$ . Integration from  $-\infty$  gives  $v \leq H$ , and hence  $H(s + \log a) \leq aH(s)$  for  $a \geq 1$ . The upper bound in (8.16) consequently shows that the radial  $\rho$  coordinates at  $\bar{\eta}$  are at most  $5/4$  times the coordinates of the first accurate  $U$  center. The proof of theorem 4.3 bounds the latter vector's norm by  $c_* L_{\text{opt},U}(\varepsilon)$ . Finally  $L_{\text{opt},U}(\varepsilon) \leq L_{\text{opt},F}(\varepsilon)$  follows from global metric domination. Apply the direct bound  $L_{\text{CP},F}(0, \bar{\eta}) \leq B_r \Gamma_k \|y(\bar{\eta})\|$  proved above, obtaining (8.10).  $\square$

#### 8.4 Radial coupling preserves the dyadic finite-sequence separation

**Corollary 8.4** (Uniform radial discrete comparison). *Use the same dyadic objective and tolerance (6.1) with any  $F_{\lambda,c}$  satisfying (8.7); the parameters may vary with  $r$ . Uniformly over this family,*

$$L_{\text{opt},F}(\varepsilon_r) = \Theta(r\sqrt{\log r}), \quad L_{\text{CP},F}(\varepsilon_r) = \Theta(r \log r).$$

*Any arbitrary-label finite sequence starting at zero, ending at actual accuracy  $\varepsilon_r$ , taking forward- $R$  moves, and lying within  $F$ -metric distance  $\delta_r$  of its labeled centers satisfies (6.5) when  $0 < R < 1$  and  $\delta_r = o(r^{2/3})$ . An explicit unrestricted route reaches accuracy in  $O_R(r\sqrt{\log r})$  moves. In particular fixed metric tubes and fixed Newton-decrement neighborhoods have  $\Theta(\sqrt{\log r})$  movement overhead, for Newton-decrement radii  $0 \leq \beta < 1/2$ .*

*Proof.* Choose an absolute  $\tau$  so that  $v(\tau + \log(4/5)) \geq 1/2$  for the standard profile. Equations (8.15) and (8.16) imply

$$y'_i(s) \geq \kappa := 7/20 \quad (s \geq a_i + \tau), \quad 0 \leq y_i(s) \leq H(s - a_i). \quad (8.17)$$

The standard center at  $T$  is accurate, so the radial center at  $T + \log(5/4)$  is accurate by (8.16). Its transformed norm is  $O(r\sqrt{\log r})$  by (6.3). The straight segment from zero to this endpoint in  $\rho$  coordinates has  $F$ -length at most  $\sqrt{11/8}$  times its  $U$ -length, by (8.11). Chord sampling gives the explicit unrestricted route. Global metric domination and the standard target-distance lower bound in theorem 6.1 give the matching  $\Omega(r\sqrt{\log r})$  distance.

For the lower bound, use thresholds  $\sigma_i = a_i + \tau$ ,  $J = \lfloor r^{2/3} \rfloor$ , cutoff  $b = \sigma_{J+1}$ , and the potential

$$P(s) = \int_{\sigma_1}^{\min(\max(s, \sigma_1), b)} \sqrt{\#\{i : \sigma_i \leq u\}} du.$$

Its total range is  $\Theta(r \log r)$ , by the same harmonic sum and bounded rounding error as in theorem 6.2. A bounded move and the two tubes give consecutive-center  $F$ -distance at most  $C_r = 2\delta_r - \log(1 - R)$ . By metric domination this also bounds the Euclidean distance between their  $y$  coordinates. All coordinates increase by (8.15), so clipping labels does not increase this distance. The first  $j$  active coordinates advance at rate at least  $\kappa$ , and successive charged thresholds are separated by at least  $\log 2$ . Thus the calculation (6.6), with  $v_0$  replaced by  $\kappa$ , bounds a progress jump by  $O(C_r \sqrt{1 + C_r})$ .

Initially, the direct prefix estimate in the proof of theorem 8.3 and (8.17) give

$$P(s_0) \leq \kappa^{-1} \int_{-\infty}^{s_0} \|y'(s)\|_2 ds \leq \kappa^{-1} \frac{25}{14} \Gamma_r \|y(s_0)\|_2 \leq \kappa^{-1} \frac{25}{14} \Gamma_r \delta_r.$$



At termination  $1 - z_{N,1} \leq \varepsilon_r$ ; scalar contraction and the terminal tube imply  $y_1(s_N) \geq T - \log r - \delta_r$ . Together with  $y_1(s) \leq H(s) \leq s_+ + 1/\sqrt{2}$ , this forces  $s_N > b$  for large  $r$ . Since  $T - b = \Theta(r^{2/3})$ , the imposed tube range is sufficient. The net progress is  $\Theta(r \log r)$  and proves the discrete bound without monotone labels. Applying the same endpoint argument to the first accurate center and integrating (8.17) gives the central-length lower bound; [theorem 8.3](#) gives its upper bound. Finally apply [lemma 6.3](#).  $\square$

**Example 8.5** (A coupling singular at the vertices). For  $r \geq 2$ , consider  $F_0 = U - \log(r - \|x\|^2)$ . It has exact parameter  $r + 1$ , so it lies outside (8.7). The sum rule gives the upper parameter; along  $x = (1 - \tau)\mathbf{1}$  its restriction is  $-(r + 1) \log \tau + O(1)$  with the corresponding first two derivative asymptotics, proving the reverse bound. It still has endpoint and same-accuracy suprema at least  $\Gamma_{r-1}$ . To verify this despite its singular coupling, set  $n = r - 1$ ,  $w_i = e^{Md_i}$  for  $i \leq n$ , and  $w_r = e^{-M^2}$ , terminating at  $s = 0$ . Stationarity gives  $0 \leq x_r(s) \leq e^{-M^2}/2$  for  $s \leq 0$ . Thus  $r - \|x\|^2 \geq 1 - x_r^2 \geq 1/2$  on this central arc and on its coordinatewise  $U$ -geodesic from zero. The extra gradient is bounded by 4 and its Hessian by  $(4 + 16r)I$  on these paths. Apply (8.3) to the first  $n$  coordinates. Their prefix intervals give length  $M\Gamma_n^2 - o_r(M)$ ; their endpoint metric coordinates are  $Md_i + O_r(1)$ , while the final coordinate is  $o(1)$ . The comparison geodesic has  $F_0$ -length at most  $M\Gamma_n + O_r(1)$ , since its Euclidean total variation is at most  $r$ . The ratio lower bound follows, and choosing the actual terminal gap gives the same-accuracy bound exactly as in [theorem 8.1](#). This example explains why a regular unused coordinate can substitute for a global facet-collar bound.

## 9 Primal movement and primal–dual completion

A primal shortcut need not admit a short dual completion. We first state classical primal–dual geometry with its metric and endpoint contract explicit, and then compare both geometries on the same dyadic LP.

### 9.1 Orthogonal speed allocation and the feasible gap set

Let  $K$  be a proper cone (closed, pointed, full dimensional), and let  $F$  be a standard self-concordant barrier satisfying  $F(tx) = F(x) - \nu \log t$  for  $t > 0$ . Use the negative-pairing conjugate

$$F_*(s) = \sup_{x \in \text{int } K} \{-\langle s, x \rangle - F(x)\}, \quad \widehat{F}(x, s) = F(x) + F_*(s).$$

Consider the strictly feasible primal–dual pair

$$\min\{\langle c, x \rangle : Ax = b, x \in K\}, \quad \max\{\langle b, y \rangle : A^*y + s = c, s \in K^*\},$$

and its exact centers  $z(\eta) = (x(\eta), s(\eta))$ , where  $s = -\eta^{-1}F'(x)$  and  $\langle x, s \rangle = \nu/\eta$ . All full primal–dual distances in this section use the Hessian of  $\widehat{F}$  on the ambient product cone; restricted feasible distances can only be larger. A dot denotes differentiation in  $\lambda = \log \eta$ .

**Proposition 9.1** (Classical speed allocation). *Put  $H = F''(x)$ ,  $g = H^{-1/2}F'(x)$ , and let  $\Pi_x$  be the Euclidean orthogonal projector onto  $H^{1/2} \ker A$ . Then*

$$\begin{aligned} \nu_P(\eta) &:= \|\dot{x}\|_{F,x}^2 = \|\Pi_x g\|^2, & \nu_D(\eta) &:= \|\dot{s}\|_{F^*,s}^2 = \|(I - \Pi_x)g\|^2, \\ \nu_P(\eta) + \nu_D(\eta) &= \nu. \end{aligned} \tag{9.1}$$

If  $A$  has full row rank, then

$$\nu_P(\eta) = \eta^2 \langle c, Mc \rangle, \quad M = H^{-1} - H^{-1}A^*(AH^{-1}A^*)^{-1}AH^{-1}. \tag{9.2}$$

Writing  $p(\eta) = \langle c, x(\eta) \rangle$  and  $d(\eta) = \langle b, y(\eta) \rangle$ , one has

$$-\eta^2 p'(\eta) = \nu_P(\eta), \quad \eta^2 d'(\eta) = \nu_D(\eta). \quad (9.3)$$

If both objectives converge to the finite optimum  $p^*$ , then

$$p(\eta) - p^* = \int_{\eta}^{\infty} \frac{\nu_P(u)}{u^2} du, \quad p^* - d(\eta) = \int_{\eta}^{\infty} \frac{\nu_D(u)}{u^2} du. \quad (9.4)$$

*Proof.* Logarithmic homogeneity and conjugacy give  $Hx = -F'(x)$ ,  $\|g\|^2 = \nu$ , and  $F''(s) = \eta^2 H^{-1}$ . Differentiating the central relation yields  $\eta \dot{s} = F'(x) - H\dot{x}$ . Thus  $v = H^{1/2} \dot{x}$  and  $d_0 = \eta H^{-1/2} \dot{s}$  satisfy  $g = v + d_0$ . Feasibility gives  $v \in H^{1/2} \ker A$  and  $d_0 \in (H^{1/2} \ker A)^{\perp}$ , proving (9.1). Removing redundant equalities leaves the feasible tangent spaces unchanged. With full row rank, differentiated primal stationarity gives  $\dot{x} = -\eta M c$ ;  $M H M = M$  proves (9.2) and the first progress identity. Differentiate  $p - d = \nu/\eta$  to obtain the second, and integrate for (9.4).  $\square$

The constant product speed is the calculation underlying Nesterov–Todd’s primal–dual theorem [21, Theorems 4.1 and 5.2]. Metric projections and conjugate Hessian isometries also appear explicitly in Ohara, Ishi, and Tsuchiya [22, Sections 4–5]. The projection calculus itself is not a contribution of this paper. Its role here is to identify exactly which part of the constant total speed is visible in a primal projection.

**Theorem 9.2** (Classical distance to feasible small-gap points). *Let  $z_0 = z(\eta_0)$  be exact central with gap  $\Delta_0 = \nu/\eta_0$ . For  $0 < \varepsilon < \Delta_0$ , let  $\mathcal{Z}_{\varepsilon}$  consist of all strictly primal–dual feasible pairs of gap at most  $\varepsilon$ . Then*

$$\sqrt{\nu/2} \log \frac{\Delta_0}{\varepsilon} \leq d_{\widehat{F}}(z_0, \mathcal{Z}_{\varepsilon}) \leq \sqrt{\nu} \log \frac{\Delta_0}{\varepsilon}. \quad (9.5)$$

*The upper bound is the length of the central route to  $z(\nu/\varepsilon)$ . Every sequence of forward  $R$ -chords,  $0 < R < 1$ , from  $z_0$  to  $\mathcal{Z}_{\varepsilon}$  therefore has at least*

$$\frac{\sqrt{\nu/2} \log(\Delta_0/\varepsilon)}{-\log(1-R)} \quad (9.6)$$

*chords, even if intermediate points violate the affine equations.*

*Proof.* At a center  $z_c = z(\eta)$ , conjugacy gives  $F'_*(s_c) = -\eta x_c$  and  $\widehat{F}(z_c) = \nu \log \eta - \nu$ . For any feasible  $z = (x, s)$ , orthogonality of primal and dual feasible differences gives  $\langle x - x_c, s - s_c \rangle = 0$ . Hence

$$D\widehat{F}(z_c)[z - z_c] = -\eta \{ \langle s_c, x - x_c \rangle + \langle x_c, s - s_c \rangle \} = -\eta \{ \text{gap}(z) - \nu/\eta \}.$$

At  $\eta = \nu/\varepsilon$ , convexity proves that  $z_c$  minimizes  $\widehat{F}$  on  $\mathcal{Z}_{\varepsilon}$ . The ambient dual norm of  $D\widehat{F}$  is exactly  $\sqrt{2\nu}$ , since each logarithmically homogeneous barrier contributes  $\nu$ . Integrating this gradient bound along any ambient curve proves the lower bound. Equation (9.1) gives the upper bound, and lemma 2.1 proves (9.6).  $\square$

This is a quantitative restatement of [21, Theorem 5.1(c), Theorem 5.2, and Corollary 5.1], which already treats the entire feasible gap set and permits intermediate affine infeasibility. It is not a new endpoint-set extension.

## 9.2 An optimal cone barrier with little primal movement

For positive block dimensions  $s_a$ , define the one-cone homogenization

$$\mathcal{H} = \{(t, y_1, \dots, y_k) : t \geq \|y_a\|_2 \ (1 \leq a \leq k)\}.$$

Its classical barrier

$$\mathcal{F}(t, y) = - \sum_{a=1}^k \log(t^2 - \|y_a\|^2) + (k-1) \log t \quad (9.7)$$

has exact logarithmic parameter  $\nu = k + 1$ . For clarity, the positive term in this formula is not justified by a subtraction rule for barriers. Place each  $y_a$  in a private column block of row  $a$  of a  $k$ -row matrix  $Y$ . Then  $YY^* = \text{diag}(\|y_a\|^2)$ , and (9.7) is the restriction of the spectral-norm-cone barrier  $(k-1) \log t - \log \det(t^2 I - YY^*)$  from [20, Proposition 5.4.6(i), p. 199]. This supplies standard self-concordance and the parameter  $k + 1$ . Restricting further to one scalar direction in each block gives the  $(k + 1)$ -dimensional infinity-norm cone; its barrier lower bound is  $k + 1$  [13, Corollary 6.2] for  $k \geq 2$ . For  $k = 1$ , its two-dimensional section is the orthant  $\mathbb{R}_+^2$ , whose sharp parameter is two. Thus the parameter is also optimal over logarithmically homogeneous barriers on  $\mathcal{H}$ .

On  $t = 1$ , the restriction is  $\sum_a b(\|y_a\|)$ , whose exact parameter is  $k$  by [proposition 3.2](#). For unit  $u_a$  and weights  $w_a \geq 0$ , minimize  $-\sum_a w_a \langle u_a, y_a \rangle$  on this slice. Its centers are  $y_a(\eta) = q(\eta w_a) u_a$ , using the scalar  $q$  of [section 4](#). Direct radial differentiation, followed by (9.1), gives

$$\nu_P(\eta) = q_{\text{eff}}(\eta) = \sum_a \left( 1 - \frac{1}{\sqrt{1 + (\eta w_a)^2}} \right), \quad \nu_D(\eta) = k + 1 - q_{\text{eff}}(\eta). \quad (9.8)$$

For  $w_1 = 1$ , all other weights zero, and  $0 < \eta_0 < \eta_1$ ,

$$L_P \leq \log(\eta_1/\eta_0), \quad L_D \geq \sqrt{k} \log(\eta_1/\eta_0).$$

The full endpoint distance is at least  $\sqrt{(k+1)/2} \log(\eta_1/\eta_0)$ , by [theorem 9.2](#). Starting the primal curve at its analytic center  $\eta = 0$  instead costs at most  $1/\sqrt{2} + \log \eta_1$  for  $\eta_1 \geq 1$ , since  $q_{\text{eff}}(\eta) \leq \eta^2/2$  on  $(0, 1]$ . Thus the ambient parameter alone gives no comparable primal lower bound.

This conclusion does not depend on a nonunique projected optimizer. Fix  $0 < \varepsilon \leq 1$ , write  $\nu = k + 1$ , and take

$$w_1 = 1, \quad w_a = \varepsilon^2/\nu^2 \ (a \geq 2), \quad \eta_0 = 1, \quad \eta_1 = \nu/\varepsilon. \quad (9.9)$$

All weights are positive, so the product-ball optimizer is uniquely  $(u_a)_a$ . The scalar inequality  $1 - (1 + z^2)^{-1/2} \leq z^2/2$  gives, throughout this interval,

$$q_{\text{eff}}(\eta) \leq 1 + \frac{\varepsilon^2}{2\nu}, \quad L_P \leq \sqrt{1 + \frac{\varepsilon^2}{2\nu}} \log \frac{\nu}{\varepsilon}, \quad L_D \geq \sqrt{k - \frac{\varepsilon^2}{2\nu}} \log \frac{\nu}{\varepsilon}.$$

The terminal full gap is exactly  $\varepsilon$ ; the primal error is at most  $\varepsilon$ . These objectives depend on the requested accuracy. For a single fixed objective with every  $w_a > 0$ ,  $q_{\text{eff}}(\eta) \rightarrow k$  eventually. The example rules out a bound uniform over objectives; it does not erase the small-weight scale from an instance-dependent asymptotic statement. The distribution and decay estimates of [section 5](#) describe this scale more precisely.

### 9.3 Three movement scales on one sparse LP

**Theorem 9.3** (The cost of a feasible dual completion). *Use exactly the dyadic  $w_i$ ,  $a_i$ ,  $T = a_r = \Theta(r)$  and  $\varepsilon_r = re^{-T}$  of (6.1). Write the box LP in the form*

$$u_i = 1 + x_i, \quad v_i = 1 - x_i, \quad u_i + v_i = 2, \quad \min \sum_i \left( -\frac{w_i}{2} u_i + \frac{w_i}{2} v_i \right). \quad (9.10)$$

The orthant barrier  $\mathcal{F} = -\sum_i (\log u_i + \log v_i)$  has parameter  $2r$  and restricts exactly to  $U(x) = \sum_i b(x_i)$ . Start at the finite full center  $z^c(0)$ , where  $s = \log \eta = 0$ . For every fixed  $0 < R < 1$ , the optimal forward  $R$ -chord counts have the following orders as  $r \rightarrow \infty$ :

| Metric and endpoint contract                                                | Number of chords           |
|-----------------------------------------------------------------------------|----------------------------|
| Restricted primal, arbitrary route from $x^c(0)$ to $x^c(T)$                | $\Theta_R(r\sqrt{\log r})$ |
| Restricted primal, exact central iterates between those endpoints           | $\Theta_R(r \log r)$       |
| Full primal-dual, any strictly feasible output of gap $\leq 2\varepsilon_r$ | $\Theta_R(r^{3/2})$        |
| Full primal-dual, exact central route from $z^c(0)$ to $z^c(T)$             | $\Theta_R(r^{3/2})$        |

The second order also holds if iterates may lie in any fixed metric tube around labeled primal centers, with arbitrary label order and actual terminal primal accuracy. The lower full bound allows intermediate affine infeasibility. The terminal point  $x^c(T)$  is  $\varepsilon_r$ -accurate, and  $z^c(T)$  has full gap exactly  $2\varepsilon_r$ .

*Proof.* Let  $\alpha_i, \beta_i$  be the dual slacks associated with  $u_i, v_i$ . The signs in (9.10) give

$$\beta_i - \alpha_i = w_i, \quad u_i \alpha_i = v_i \beta_i = e^{-s}. \quad (9.11)$$

Consequently  $x_i^c(s) = q(e^{s-a_i})$  and

$$\|\dot{z}_P^c\|^2 = q_{\text{eff}}(e^s), \quad \|\dot{z}_D^c\|^2 = 2r - q_{\text{eff}}(e^s) \geq r, \quad \|\dot{z}^c\|^2 = 2r. \quad (9.12)$$

The full central length on  $[0, T]$  is exactly  $\sqrt{2r} T$ . The scalar error bound  $w[1 - q(\eta w)] \leq 1/\eta$  proves terminal primal accuracy; summing (9.11) gives full gap  $2re^{-T} = 2\varepsilon_r$ .

Here is a direct lower bound to every feasible full target. Initially  $1/\alpha_i^0 + 1/(\alpha_i^0 + w_i) = 2$ ; as  $w_i \leq 1$ , the decreasing left side implies  $\alpha_i^0 \geq 1/\sqrt{2}$ . At every feasible target,

$$\text{gap} = \sum_i (u_i \alpha_i + v_i \beta_i) = \sum_i (2\alpha_i + v_i w_i).$$

Thus  $\text{gap} \leq 2\varepsilon_r$  forces every  $\alpha_i \leq \varepsilon_r$ . The dual orthant metric is Euclidean in coordinatewise logarithms. Projecting an arbitrary full curve onto these  $r \log$ -slacks proves

$$d_{\widehat{\mathcal{F}}}(z^c(0), \mathcal{Z}_{2\varepsilon_r}) \geq \sqrt{r} [T - \log r - \tfrac{1}{2} \log 2]_+. \quad (9.13)$$

Together with the central upper route and lemma 2.1, this proves both full rows, with no prescribed final dual certificate. Theorem 9.2 supplies another lower proof of the same order.

For the primal rows, theorem 6.1 starts at the analytic center and ends at the first accurate parameter  $s_\varepsilon \in (T-1, T]$ . Its negative-parameter tail has length  $O(1)$ : the scalar bound  $v(s-a_i) \leq e^{s-a_i}/\sqrt{2}$  bounds its length by  $(\sum_i w_i^2/2)^{1/2} = O(1)$ . The last segment from  $s_\varepsilon$  to  $T$  has length at most  $\sqrt{r}$ . Triangle inequalities therefore preserve the primal endpoint distance  $\Theta(r\sqrt{\log r})$  and central length  $\Theta(r \log r)$ . The endpoint distance and chord sampling prove the arbitrary-route row and the central upper bound. For the central lower bound, prepend a bounded number of sampled chords on the negative-parameter tail and apply the discrete potential theorem, theorem 6.2. This argument also proves the fixed-tube statement and allows backward labels. It uses the discrete potential, not central arclength alone.  $\square$

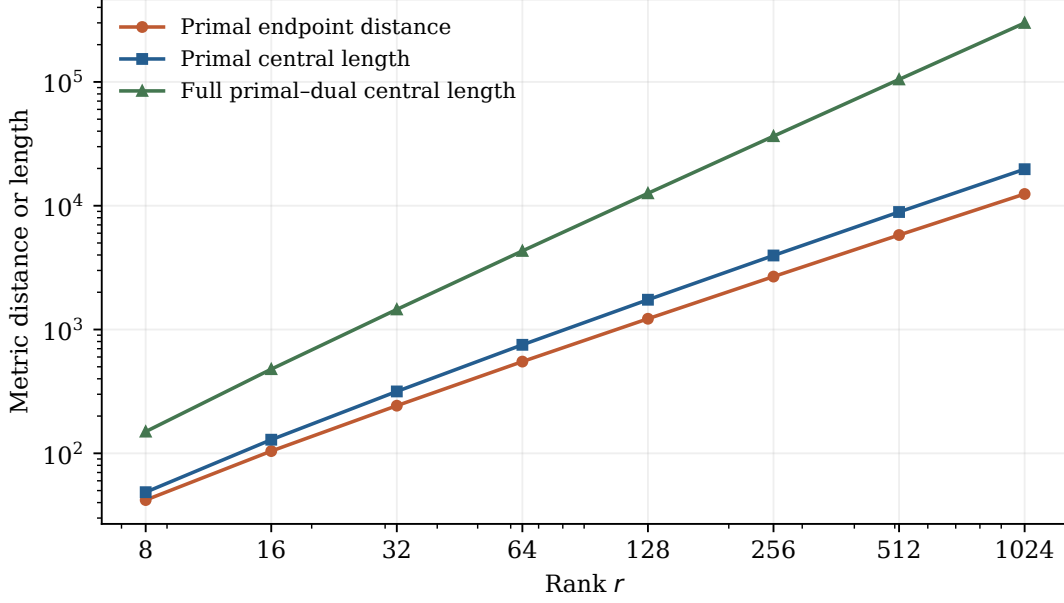


Figure 2: Numerical values on the exact dyadic family (6.1), with finite initial logarithmic parameter  $s = 0$  and terminal parameter  $T = a_r$ . The curves show the exact-formula primal endpoint distance  $d_U(x^c(0), x^c(T))$ , the primal central arclength on  $[0, T]$  evaluated by quadrature, and the full central length  $\sqrt{2r} T$ . The first curve is not the minimum distance to the entire accurate target set, and the third is not the minimum full gap-set distance. The respective orders  $r\sqrt{\log r}$ ,  $r \log r$ , and  $r^{3/2}$ , and the matched movement comparisons, follow from theorems 6.1 and 9.3. Data and generation code accompany the paper.

The ordinary rational encoding length of these same data is  $B = \Theta(r^2)$  by theorem 6.1. Thus the three scales are

$$\Theta(\sqrt{B \log B}), \quad \Theta(\sqrt{B} \log B), \quad \Theta(B^{3/4}). \quad (9.14)$$

A feasible dual certificate in this metric increases the best movement by a factor  $\Theta(\sqrt{r/\log r})$  relative to the best primal route. Each equality row has two nonzeros and each column one; sparsity does not remove this geometric cost. This comparison concerns bounded local movement. Neither this theorem nor the speed identity is an unconditional running-time or oracle-query lower bound.

## 10 What a change of formulation can preserve

The completion theorem fixes a formulation as well as a barrier. This section gives two ways to obtain movement bounds for other formulations. Exposed support rank controls primal movement for standard symmetric-cone metrics; a dimension cap controls full primal-dual movement for arbitrary logarithmically homogeneous barriers. Neither statement infers a primal lower bound from a parameter upper bound.

### 10.1 Exposed support rank and arbitrary auxiliary motion

Let  $K = \prod_i K_i$  be a product of symmetric cones in Euclidean Jordan algebras with trace inner products, and let  $\Omega = \{x \in \text{int } K : Ax = b\} \neq \emptyset$ . Suppose a support objective has finite optimum

$\ell_*$  and an attained dual certificate  $s = (s_i) \in K$  satisfying

$$\ell_* - \ell(x) = \sum_i \langle s_i, x_i \rangle \quad (x \in \Omega). \quad (10.1)$$

Let  $c_i$  be the support idempotent of  $s_i$ , let  $q_i = \text{rank}(s_i)$ , and omit zero-rank factors from expressions below. Here  $P(c_i)$  is the orthogonal Peirce projection onto the algebra with unit  $c_i$ , and  $\det_{c_i}$  is its determinant.

**Theorem 10.1** (Distance forced by the exposed minor). *Use the restricted barrier*

$$F(x) = - \sum_i \alpha_i \log \det_{c_i} x_i, \quad \alpha_i \geq 1.$$

Put  $Q_\alpha = \sum_i \alpha_i q_i > 0$  and, for any reference  $x^c \in \Omega$ , define

$$\Delta_{c,\alpha} = Q_\alpha \left[ \prod_i \frac{\{\det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^c)\}^{\alpha_i}}{\alpha_i^{\alpha_i q_i}} \right]^{1/Q_\alpha}. \quad (10.2)$$

For  $\varepsilon > 0$ , every feasible  $y$  of objective gap at most  $\varepsilon$  satisfies

$$d_F(x^c, y) \geq \sqrt{Q_\alpha} [\log(\Delta_{c,\alpha}/\varepsilon)]_+. \quad (10.3)$$

There is no restriction on unexposed eigenvalues or auxiliary fibers, and strict complementarity is not required. If  $d_F(x_0, x^c) \leq D_0$  and each of  $N$  counted rounds has metric displacement at most  $B_0 > 0$ , then

$$N \geq \frac{[\sqrt{Q_\alpha} \log(\Delta_{c,\alpha}/\varepsilon) - D_0]_+}{B_0}. \quad (10.4)$$

In particular, an initial local error  $\|x_0 - x^c\|_{x^c} \leq \rho$ ,  $0 \leq \rho < 1$  and at most  $m \in \mathbb{N}$  forward  $R$ -chords per round,  $0 < R < 1$ , allow  $D_0 = -\log(1 - \rho)$  and  $B_0 = -m \log(1 - R)$ .

*Proof.* We give the contraction argument to specify the metric normalization, including the exceptional Jordan algebra. For one idempotent  $c$  of rank  $q$ , put  $a = P(c)x$  and  $\phi_c(x) = -\log \det_c a$ . The Hessian of  $-\log \det x$  is  $P(x^{-1})$  with inverse  $P(x)$ , where  $P(u) = 2L_u^2 - L_{u^2}$  is the quadratic representation and  $L_u v = u \circ v$ . Since  $a^{-1}$  belongs to the Peirce algebra,

$$d\phi_c(x)[h] = -\langle a^{-1}, h \rangle, \quad \|d\phi_c(x)\|_{x,*}^2 = \langle a^{-1}, P(x)a^{-1} \rangle = q.$$

Indeed, the fundamental identity  $P(P(u)v) = P(u)P(v)P(u)$  with  $u = c$  gives  $P(c)P(x)P(c) = P(a)$ ; then  $P(a)a^{-1} = a$  and  $\langle a^{-1}, a \rangle = q$ . The compression  $a$  is positive definite in its Peirce algebra: if  $x \succeq \delta e$ , positivity of  $P(c)$  gives  $a \succeq \delta c$ . For  $\Phi = \sum_i \alpha_i \phi_{c_i}$ , the product metric therefore gives  $\|d\Phi\|_{F,*}^2 = Q_\alpha$  before affine restriction and at most  $Q_\alpha$  afterwards. These quadratic-representation identities are standard symmetric-cone algebra [10]; the principal-minor metric identity is not asserted as new.

Set  $g_i = \langle s_i, y_i \rangle > 0$  on the active factors. Inside the Peirce algebra,  $z_i = P(s_i^{1/2})P(c_i)y_i$  has trace  $g_i$  and determinant  $\det_{c_i}(s_i) \det_{c_i}(P(c_i)y_i)$ . Spectral AM–GM gives

$$\det_{c_i}(P(c_i)y_i) \leq \det_{c_i}(s_i)^{-1} (g_i/q_i)^{q_i}.$$

Under  $\sum_i g_i \leq \varepsilon$ , the weighted product  $\prod_i (g_i/q_i)^{\alpha_i q_i}$  is maximized at  $g_i = \varepsilon \alpha_i q_i / Q_\alpha$ . Thus

$$\prod_i \det_{c_i}(P(c_i)y_i)^{\alpha_i} \leq (\varepsilon/Q_\alpha)^{Q_\alpha} \prod_i \alpha_i^{\alpha_i q_i} \det_{c_i}(s_i)^{-\alpha_i}.$$

Comparing with  $x^c$  proves  $\Phi(y) - \Phi(x^c) \geq Q_\alpha \log(\Delta_{c,\alpha}/\varepsilon)$ . Integrate the covector norm bound along every feasible curve. The triangle inequality and lemma 2.1 prove the round statements.  $\square$

For unit scales this gives  $Q = \sum_i q_i$  and  $\Delta_c = Q[\prod_i \det_{c_i}(s_i) \det_{c_i}(P(c_i)x_i^e)]^{1/Q}$ . The coefficient  $\sqrt{Q_\alpha}$  is asymptotically sharp on the full cone. A quadratic-representation isometry sends the reference to the unit; diagonalize the transformed certificate and multiply its support coordinates by  $t$ , keeping all other coordinates one. The gap is  $t\langle s, e \rangle$  and the curve has length  $\sqrt{Q_\alpha} \log(1/t)$ . Together with (10.3), this proves the exact leading coefficient as  $\varepsilon \downarrow 0$ . An affine slice may exclude this upper path; no sharp upper bound on an arbitrary slice is claimed.

Hauser and Güler's classification [12, Theorem 5.5] states that every self-scaled barrier on a symmetric cone is, up to an additive constant, a sum of these log-determinants on its irreducible factors with coefficients  $\alpha_i \geq 1$ . Thus theorem 10.1 covers every such barrier, with its actual weighted scale; simply replacing  $Q$  by  $Q_\alpha$  while retaining the unweighted  $\Delta_c$  would be incorrect. The theorem is conditional on the certificate rank. A general lower bound on the ranks attainable by different lift dictionaries is a separate question, not an assumption hidden in this distance estimate.

## 10.2 A dimension-only consequence for arbitrary cone barriers

**Theorem 10.2** (Bounded operational factor dimension). *Let a compact full-dimensional convex body  $C \subset \mathbb{R}^D$ ,  $D \geq 1$ , have an exact affine lift  $C = \pi(K \cap \mathcal{L})$ , where  $K$  is a finite product of proper cones,  $\mathcal{L}$  is an affine subspace, and  $\pi$  is an affine map. The minimal operational face is the smallest face of  $K$  containing the feasible lifted slice  $K \cap \mathcal{L}$ ; it is a product of faces of the original factors. Regard this product in its linear span and remove its zero factors to obtain the operational product cone. Suppose each factor of this reduced cone has dimension at most an integer  $d \geq 2$ . For  $h \geq 1$  define*

$$\Psi_d(h) = 2\lfloor h/d \rfloor + \min\{h \bmod d, 2\}.$$

*Every logarithmically homogeneous standard self-concordant barrier on that operational product cone, including a coupled barrier, satisfies*

$$\nu \geq \Psi_d(D+1) \geq \frac{2(D+1)}{d}. \quad (10.5)$$

*For a strictly feasible conic pair on that cone, an exact central start of gap  $\Delta_0$ , a strictly feasible terminal gap at most  $0 < \varepsilon < \Delta_0$ , and at most  $m$  forward product-metric  $R$ -chords per round, where  $m \in \mathbb{N}$  and  $0 < R < 1$ , the number of rounds is at least*

$$\frac{\sqrt{\Psi_d(D+1)/2} \log(\Delta_0/\varepsilon)}{-m \log(1-R)}. \quad (10.6)$$

*Proof.* Let  $M$  be the dimension of the operational product space. The affine projection of its feasible slice spans  $\mathbb{R}^D$ , so  $M \geq D$ . If  $M = D$ , its linear projection is invertible and the slice has full affine hull in the operational space. There can then be no effective affine equality restriction: the feasible slice is the whole cone. Its invertible affine image is unbounded, contrary to compactness. Therefore  $M \geq D+1$ . This elementary dimension argument is compatible with the classical slack-factorization framework [11], but needs no selection of boundary certificates.

Let  $r_0$  count ray factors and  $q_0$  count the factors of dimension at least two. In each nonray factor choose a two-dimensional linear plane containing an interior point. Its cone section is proper and full-dimensional in the plane, hence linearly isomorphic to  $\mathbb{R}_+^2$ ; its relative interior lies in the factor's interior and its boundary in the factor's boundary. Together with all rays, these sections give an interior orthant section of dimension  $r_0 + 2q_0$ . Restriction preserves standard self-concordance, logarithmic degree, and barrier blow-up. The classical orthant parameter lower bound [13, 20] therefore gives  $\nu \geq r_0 + 2q_0$ , even when the original barrier couples factors. The dimension cap gives  $r_0 + dq_0 \geq M \geq D+1$ .



To solve the integer minimization, write  $h = ad + b$ ,  $0 \leq b < d$ . For  $q_0 \leq a$  the smallest possible charge is  $2q_0 + h - dq_0 = h - (d - 2)q_0$ , minimized at  $q_0 = a$  and equal to  $2a + b$ . For  $q_0 \geq a + 1$  the charge is at least  $2a + 2$ , attained with no rays. The minimum is therefore  $2a + \min(b, 2)$ ; the case  $b = 0$  is attained with  $q_0 = a$ . Since  $\min(b, 2) \geq 2b/d$ , this proves (10.5). Apply theorem 9.2 for (10.6).  $\square$

The parameter conclusion concerns a barrier on the operational cone, not an arbitrary barrier on the projected body or affine slice. This is a short synthesis of classical restriction, dimension, and primal–dual geometry. It does not assert an optimal lift-dimension frontier. For a product of balls,  $D = \sum_a s_a$ ; the bound therefore sees the total source dimension without curvature or regularity hypotheses.

### 10.3 Grouped balls, norm trees, and PSD completion

The following concrete formulations illustrate why reducing the number of conic blocks need not reduce movement in the displayed restricted barrier. Full proofs, including the treatment of all auxiliary variables, are in appendix C.

Take  $k$  unit balls, a unit support direction  $c_a$  for each, and objective  $\ell = \sum_a \langle c_a, w_a \rangle$  of optimum  $k$ . For a grouped paraboloid lift, partition each ball into  $h$  coordinate groups, introduce  $q_{aG} = s_{aG} - \|w_{aG}\|^2 > 0$  with  $\sum_G s_{aG} = 1$ , and use  $F = -\sum_{a,G} \log q_{aG}$ . For a PSD packing, place these  $H = kh$  group columns into arbitrary real matrix blocks  $W_\ell$ , allow every off-diagonal completion variable in  $S_\ell$ , and use

$$D_\ell = S_\ell - W_\ell^T W_\ell \succ 0, \quad F = -\sum_\ell \log \det D_\ell, \quad \sum_{\gamma \text{ from ball } a} (S_{\ell(\gamma)})_{\gamma\gamma} = 1.$$

Padding a group column with zeros is allowed; its free entries are its own group coordinates. Each group occurs exactly once. Both barriers have standard self-concordance and exact restricted parameter  $H$ . Their central path  $z(\tau) = \arg \min(F - \tau\ell)$  has  $w_a = r(\tau)c_a$ , with

$$q(\tau) = \frac{2}{\sqrt{h^2 + \tau^2} + h}, \quad r(\tau) = \frac{\tau}{\sqrt{h^2 + \tau^2} + h}, \quad \left\| \frac{dz}{d \log \tau} \right\|^2 = H \left( 1 - \frac{h}{\sqrt{h^2 + \tau^2}} \right). \quad (10.7)$$

The residuals are  $q_{aG} = q(\tau)$  or  $D_\ell = q(\tau)I$ . For every  $\tau_0 \geq 0$  and every accurate feasible  $y$ ,

$$d_F(z(\tau_0), y) \geq \sqrt{H} [\log(q(\tau_0)H/(2\varepsilon))]_+. \quad (10.8)$$

In particular the analytic-center logarithmic scale is  $k/(2\varepsilon)$ , independent of the packing and its off-diagonal variables.

For a rooted norm tree with  $b$  internal nodes, each with at least two children, use one positive Lorentz block per node:

$$q_{av} = t_{av}^2 - \sum_{u \in I(v)} t_{au}^2 - \sum_{j \in J(v)} w_{aj}^2 > 0, \quad t_{av} > 0, \quad t_{a,\text{root}} = 1. \quad (10.9)$$

Here  $I(v)$  and  $J(v)$  denote internal and leaf children. The positive sheet is essential for convexity. Use  $F = -\sum_{a,v} \log q_{av}$  and put  $L = kb$ . Its exact parameter is  $\nu = k\nu_T$ , where

$$\nu_T = \begin{cases} 2b - 2, & \text{the root has only internal children,} \\ 2b - 1, & \text{the root has a leaf child.} \end{cases} \quad (10.10)$$

Its central path is (10.7) with  $h = b, H = L$ , regardless of tree shape. If  $n_v$  counts internal nodes below  $v$  including  $v$ , and  $C_{av} = \sum_{j \text{ below } v} c_{aj}^2$ , the internal coordinates are  $t_{av}^2 = n_v q + r^2 C_{av}$ . At the analytic center  $\bar{z}$ ,  $q_{av} = 1/b$ , and every accurate  $y$  satisfies the sharper barrier-height bound

$$d_F(\bar{z}, y) \geq \frac{L}{\sqrt{\nu}} [\log(k/(2\varepsilon))]_+ \geq \sqrt{L/2} [\log(k/(2\varepsilon))]_+. \quad (10.11)$$

The exact parameter proof is included because subtracting the fixed root's contribution without a metric calculation would be unjustified.

For all three formulations the central route to error  $\varepsilon$ , from the analytic center, has length

$$\sqrt{M_0} \rho(1 - \varepsilon/k) = \sqrt{M_0} \log(k/\varepsilon) + O(\sqrt{M_0}), \quad M_0 = H \text{ or } L, \quad (10.12)$$

for  $0 < \varepsilon < k$ , with the asymptotic as  $\varepsilon/k \downarrow 0$ . Thus sampling this route and applying the distance lower bounds gives matching  $\Theta_R(\sqrt{M_0} \log(k/\varepsilon))$  chord counts, uniformly for  $0 < \varepsilon \leq k/4$ . Approximate initialization and a fixed number of chords per round are handled exactly as in (10.4). These are statements about the specified metrics; block count alone would prove none of these primal lower bounds.

## 11 Interpretation

The comparisons identify three distinct sources of movement. Objective weights determine which directions become active and at what parameter scales. Ordered activation can make the central route longer than a simultaneous motion toward an accurate endpoint. A feasible dual completion can then require additional movement that is absent from the primal projection. On the dyadic cube family these effects have separate, matched asymptotic orders in both length and bounded local moves.

The exact target allocation is useful because a central endpoint need not be the closest accurate point. It also shows why one aggregate rank or determinant certificate may be insufficient: the entire objective weight profile enters the optimum. Conversely, the sharp prefix factor shows that ordered scalar activation imposes a strong restriction on how inefficient a central route can be. For equal standard scales the worst overhead is logarithmic in active rank under a square root, even when the ambient matrix space is much larger.

Changing a barrier or a formulation requires a new metric comparison. Optimal parameter alone does not eliminate the centrality cost, as the canonical cube barriers and the explicit coupled radial family show. A restriction of a cone barrier can also have much smaller primal activity than the full homogeneous parameter suggests. The exposed-rank and concrete-formulation results provide direct lower certificates when auxiliary motion is permitted, rather than inferring primal cost from an ambient parameter bound.

All these conclusions retain their stated contracts: fixed barrier metric, specified starting point, actual terminal accuracy or feasible duality gap, and bounded starting local norms. Within those contracts, the explicit shortest routes and matching lower bounds make the cost of centrality and the cost of dual completion separately measurable.

## A Rational certificates for the scalar dilation

This appendix proves the strict bounds in (4.9). All finite arithmetic checks below are reproduced by the standalone standard-library script `scripts/verify_scalar_certificate.py`; no floating-point evaluation is used by the certificate.

## A.1 An elasticity bound

For  $x = \rho^{-1}(y) > 0$ , define

$$E(y) = \frac{yp'(y)}{p(y)}, \quad K(y) = \frac{y^2(1-x^2)}{2x^2(1+x^2)}.$$

Direct differentiation using  $p = b' \circ \rho^{-1}$  gives

$$\frac{dE}{d \log y} = E - K, \quad E \geq 1, \quad 0 < K < 1. \quad (\text{A.1})$$

For the last inequality, set  $B(x) = \sqrt{2x}\sqrt{(1+x^2)/(1-x^2)}$ , so  $K = (\rho(x)/B(x))^2$ . With  $u = x^2$ , the derivative ratio is

$$\frac{B'}{\rho'} = \frac{1+2u-u^2}{(1+u)\sqrt{1-u}} > 1,$$

because the difference of squared numerator and denominator is  $u(3+3u-3u^2+u^3) > 0$ . Equality of the two functions at zero proves  $\rho < B$ . Convexity of  $p$  gives  $E \geq 1$ , so  $E$  increases strictly. Solving the differential inequality in (A.1) on a multiplicative ray gives  $E(ty) > 1 + t(E(y) - 1)$  for  $t > 1$ . Consequently, for every  $a > 1$ ,

$$\log \frac{p(ay)}{yp'(y)} > h_1(E) := \log a + (a-1)(E-1) - \log E. \quad (\text{A.2})$$

For a stronger bound when  $E > 2$ , concavity of the square root gives

$$\rho(x) \leq \sqrt{2}D(x), \quad D(x) = \frac{3}{2} \operatorname{artanh} x - x/2.$$

At  $x = 4/5$ ,  $D = (3/2) \log 3 - 2/5 < 2497/2000$ . Indeed  $\log 3 < 1099/1000$  follows by summing the exponential Taylor polynomial through degree six. Hence  $E(\rho(4/5)) < 2497\sqrt{41}/8000 < 2$ , the latter inequality being  $2497^2 \cdot 41 < 64 \cdot 2000^2$ . Thus  $E(y) > 2$  implies  $x > 4/5$ . On this region,

$$K \leq G(x) := \frac{D(x)^2(1-x^2)}{x^2(1+x^2)}.$$

The positive power series for  $\operatorname{artanh}$  gives  $D(x) \geq x(1+x^2/2)$ , and  $D'(x) = (1+x^2/2)/(1-x^2)$ . It follows that

$$\frac{1}{2}(\log G)' = \frac{D'}{D} - \frac{x}{1-x^2} - \frac{1}{x} - \frac{x}{1+x^2} \leq -\frac{x}{1+x^2} < 0.$$

Therefore

$$K < G(4/5) < \left(\frac{2497}{2000}\right)^2 \frac{225}{656} = \frac{56115081}{104960000} < k := \frac{107}{200}.$$

This bound holds along the entire multiplicative ray starting at  $E > 2$ . Integrating (A.1) now gives  $E(ty) > k + t(E(y) - k)$  for  $t > 1$ , and hence

$$\log \frac{p(ay)}{yp'(y)} > h_k(E) := k \log a + (a-1)(E-k) - \log E \quad (E > 2). \quad (\text{A.3})$$

## A.2 The strict upper certificate

Take  $a = 69/50$ . On  $1 \leq E \leq 2$ ,  $h_1$  is decreasing, and

$$h_1(E) \geq h_1(2) = \log(69/100) + 19/50 > 0,$$

because  $e^{19/50} > 1 + 19/50 + (19/50)^2/2 = 7261/5000 > 100/69$ . On  $E > 2$ , the convex function  $h_k$  has its global minimum at  $E = 50/19$ , where its value is

$$\log(19/50) + 1 + \frac{107}{200}(\log(69/50) - 19/50).$$

The positive series  $\log((1+z)/(1-z)) = 2\sum_{j\geq 0} z^{2j+1}/(2j+1)$  yields, with  $r_0 = 31/69$  and  $s_0 = 19/119$ ,

$$\begin{aligned}\log(19/50) &> -2 \left( r_0 + \frac{r_0^3}{3} + \frac{r_0^5}{5(1-r_0^2)} \right) = -\frac{9064603453}{9362506500}, \\ \log(69/50) &> 2(s_0 + s_0^3/3) = \frac{1628072}{5055477}.\end{aligned}$$

Substitution leaves the strictly positive rational lower bound

$$h_k(50/19) > \frac{255839420916449}{315546241820670000} > 0. \quad (\text{A.4})$$

Thus  $p((69/50)y) > yp'(y)$  for every  $y > 0$ . Since the maximum defining  $c_\star$  is attained, the upper bound on  $c_\star$  is strict.

### A.3 The strict lower certificate

The elementary transform in (4.11) has series

$$Y(z) = \sum_{j \geq 0} \frac{(2 - 2^{-j})z^{2j+1}}{2j+1}, \quad S_N(z) < Y(z) < S_N(z) + \frac{2z^{2N+3}}{(2N+3)(1-z^2)}. \quad (\text{A.5})$$

Here  $S_N$  includes  $j = 0, \dots, N$ ; the tail estimate follows by bounding  $2 - 2^{-j} < 2$  and every tail denominator by  $2N + 3$ . Set

$$v_0 = \frac{3787}{4000}, \quad w_0 = \frac{9797031}{10^7}, \quad a_0 = \frac{68743}{50000}.$$

Summation through  $N = 1200$  and rational squaring for  $A$  verify

$$\frac{2453756741730599}{10^{15}} < Y(v_0) < \frac{12268783708653}{5 \cdot 10^{12}}, \quad \frac{134943211257327}{4 \cdot 10^{13}} < Y(w_0),$$

$$A(v_0) > \frac{405367307458211}{4 \cdot 10^{13}}, \quad A(w_0) < \frac{25381942601625079}{10^{15}}.$$

These finite enclosures imply

[illegible]

The second inequality states  $P(w_0) < Y(v_0)A(v_0)$ . Since  $P$  and  $Y$  are strictly increasing, it follows that  $Y(W(v_0)) > Y(w_0) > a_0 Y(v_0)$ . This is a strictly feasible witness for  $c_\star > a_0$ . The interval bounds, tail estimate, and rational squaring are a finite proof certificate, independent of a numerical optimizer or an assumption about stationary-point uniqueness.

#### A.4 Every scalar-dilation maximizer lies in a certified interval

The same inequalities also locate every maximizer of the distinct standard-barrier constant  $c_*$ . Keep  $a_0 = 68743/50000$  and  $k = 107/200$  in [equations \(A.2\)](#) and [\(A.3\)](#). Exact positive-series bounds for logarithms give

$$\begin{aligned} h_1(2) &> 2589087717927/4000000000000000 > 0, \\ h_k(571/250) &> 2011885403423/10000000000000000 > 0, \\ h_k(773/250) &> 8579961311243/50000000000000000 > 0. \end{aligned}$$

For clarity these are finite certificates: for rational  $q \in (0, 1)$ , the partial sum of  $2 \sum_{j \geq 0} q^{2j+1}/(2j+1)$  through  $j = N$  bounds  $\log((1+q)/(1-q))$  below, and adding  $2q^{2N+3}/[(2N+3)(1-q^2)]$  bounds it above. Taking  $q = (z-1)/(z+1)$  for  $z = a_0, 2, 571/250, 773/250$  and  $N = 200$  verifies the displayed inequalities using rational arithmetic. The function  $h_1$  decreases on  $[1, 2]$ . The convex function  $h_k$  decreases until  $E = 1/(a_0 - 1)$  and increases thereafter; this minimum lies strictly between  $571/250$  and  $773/250$ . Therefore a scalar ratio greater than  $a_0$  is possible only for  $571/250 < E < 773/250$ .

In the coordinate of [\(4.11\)](#),  $E = Y(v)/v$  is strictly increasing by [\(A.1\)](#). The series [\(A.5\)](#) through  $N = 1200$  gives

$$\begin{aligned} \frac{571}{250} \frac{4611}{5000} - Y(4611/5000) &> 159652343530451/20000000000000000 > 0, \\ Y(9701/10000) - \frac{773}{250} \frac{9701}{10000} &> 19492543073837/25000000000000000 > 0. \end{aligned}$$

Since every global maximizer has ratio  $c_* > a_0$ , these inequalities, followed by rational squaring in  $x = v/\sqrt{2-v^2}$ , prove

$$\frac{4611}{5000} < v_* < \frac{9701}{10000}, \quad \frac{43}{50} < x_* < \frac{943}{1000}. \quad (\text{A.7})$$

The verification script includes every displayed bound. Finally, differentiating  $P(W(v)) = Y(v)A(v)$  shows that every stationary point satisfies

$$Y(v)[Y'(v)A(v) + Y(v)A'(v)] = Y(W(v))Y'(v)A(W(v)), \quad A'(v) = \frac{v(3-v^2)}{(1-v^2)^2\sqrt{2-v^2}}.$$

No uniqueness of this stationary point or of the  $c_*$  maximizer is inferred from its certified localization.

## B The scalar envelope and the nonmonotone construction

**Proposition B.1** (Exact relaxed envelope at the safe scale). *Let  $p \in C^2([0, \infty))$  satisfy  $p(0) = 0$ ,  $p' > 0$ ,  $|p''| \leq p'$ , and  $p \leq p'$  on  $[0, \infty)$ . Put  $k = \log 2$  and*

$$C(a) = \begin{cases} 1 + \frac{1}{a} \log \frac{k}{k(2-e^{-a})-a}, & 2a \leq k(2-e^{-a}), \\ 1 + \frac{1}{a} \log \frac{4a}{k(2-e^{-a})^2}, & 2a \geq k(2-e^{-a}). \end{cases} \quad (\text{B.1})$$

Then  $C_{\text{sc}} = \max_{a>0} C(a)$  has a unique maximizer, is strictly below two, and

$$a \leq p^{-1}(ap'(a)/k) \leq C(a)a \leq C_{\text{sc}}a.$$

The function  $C(a)$  is the exact envelope obtained in the relaxed class of locally absolutely continuous functions  $v : [0, \infty) \rightarrow [0, 1]$  satisfying

$$v(0) = 0, \quad 0 < v(y) \leq 1 \quad (y > 0), \quad 1 - v \leq v' \leq 1 + v \quad a.e. \quad (\text{B.2})$$

For the original  $C^2$  function  $p$ , this profile is  $v = p/p'$ . This sharpness is at the fixed scale  $k$ , and is not a claim about the optimal arclength constant for products of one smooth barrier.

*Proof.* The derivative identity  $v' = 1 - vp''/p'$  gives (B.2). Integration yields

$$1 - e^{-a} \leq v(a) \leq \min(1, e^a - 1), \quad \frac{v(a)}{a} \leq \frac{1}{k}.$$

For the last inequality,  $(e^a - 1)/a$  increases and  $1/a$  decreases; their crossing is at  $a = k$ . Thus the comparison point  $a + t$ , defined by  $p(a + t) = ap'(a)/k$ , has  $t \geq 0$ . Write  $u = v(a)$  and integrate  $(\log p)' = 1/v$ :

$$\int_0^t \frac{dq}{v(a+q)} = \log \frac{a}{ku}. \quad (\text{B.3})$$

The upper differential constraint and the cap give  $v(a+q) \leq m_u(q) := \min\{1, (1+u)e^q - 1\}$ . The cap is reached at  $q_u = \log(2/(1+u))$ . Before then,

$$\int_0^t \frac{dq}{m_u(q)} = \log \frac{(1+u) - e^{-t}}{u}.$$

After then it equals  $\log((1+u)/(2u)) + t - q_u$ . The right side of (B.3) is fixed, and the actual integral is at least this envelope integral. Solving the latter therefore gives an upper bound on  $t$ :

$$t \leq \begin{cases} \log \frac{k}{k(1+u) - a}, & 2a \leq k(1+u), \\ \log \frac{4a}{k(1+u)^2}, & 2a \geq k(1+u). \end{cases}$$

Both expressions decrease with  $u$ , and their values and first derivatives agree at the switch. Substitute  $u \geq 1 - e^{-a}$  to obtain (B.1).

There is one positive branch boundary:  $2a - k(2 - e^{-a})$  has positive derivative  $2 - ke^{-a}$  and changes sign. It lies below  $k$ . On the first branch,  $C(a) < 2$  is equivalent to  $2k(1 - e^{-a}) - a > 0$ . This strictly concave function vanishes at 0 and  $k$ , and the branch has  $0 < a < k$ . On the second branch,  $C(a) < 2$  is equivalent to

$$k(4e^a - 4 + e^{-a}) - 4a > 0.$$

Since  $k > 2/3$ , the expression exceeds  $(2/3)(4e^a + e^{-a} - 4 - 6a)$ . The inequalities  $e^a \geq 1 + a + a^2/2 + a^3/6$  and  $e^{-a} \geq 1 - a$  bound the parenthesis below by  $1 - 3a + 2a^2 + (2/3)a^3$ . Its minimum on  $[0, \infty)$  is  $(16 - 5\sqrt{10})/3 > 0$ , as differentiation shows. Here  $k > 2/3$  follows, for example, from the first term of the positive series for  $\log 2 = 2 \operatorname{artanh}(1/3)$ .

The two branches are continuously differentiable at their common boundary. Their endpoint limits are  $1/k$  as  $a \downarrow 0$  and 1 as  $a \rightarrow \infty$ . Moreover  $C(k) = 1 + k^{-1} \log(16/9) > 1/k$ , because  $\log(32/9) > 1$  (use  $e < 3 < 32/9$ ). Hence continuity proves a maximum at a finite positive argument, and the strict pointwise bound implies  $C_{\text{sc}} < 2$ . For direct numerical evaluation and uniqueness, write  $C = 1 + B/a$ , where  $B$  is the logarithm on the appropriate branch. An interior stationary point satisfies  $aB' = B$ , with

$$B' = \frac{1 - ke^{-a}}{k(2 - e^{-a}) - a} \quad \text{or} \quad B' = \frac{1}{a} - \frac{2e^{-a}}{2 - e^{-a}}.$$

On the first branch,  $B = -\log(2 - e^{-a} - a/k)$  is strictly convex, since its positive logarithm argument has second derivative  $-e^{-a} < 0$ . As  $B(0) = 0$ ,  $aB' - B > 0$  there. On the second,

$$B'' = -\frac{1}{a^2} + \frac{4e^{-a}}{(2 - e^{-a})^2} < 0.$$

Indeed the strict inequality is equivalent to  $2e^{a/2} - e^{-a/2} > 2a$ ; the Taylor bounds  $e^t \geq 1 + t + t^2/2$  and  $e^{-t} \leq 1 - t + t^2/2$  for  $t \geq 0$  give a difference at least  $1 - a/2 + a^2/8 > 0$ . Thus  $aB' - B$  strictly decreases on the second branch, starts positive by continuous differentiability at the switch, and tends to  $-\infty$ . It has one zero, which is the unique global maximizer. Numerical evaluation gives  $a \approx 0.65011438345$  and  $C_{\text{sc}} \approx 1.831856423$ ; these decimals are not used in the bound or in its attainment proof.

Finally, for fixed  $a$  take  $v(y) = 1 - e^{-y}$  until  $a$ , then follow  $v' = 1 + v$  until reaching one, and remain one thereafter. This continuous, piecewise differentiable function obeys (B.2) almost everywhere and attains every envelope equality used at  $a$ . Reconstruct  $p$  by  $(\log p)' = 1/v$ , with normalization  $p'(0) = 1$ ; the behavior  $v(y) = y + O(y^2)$  gives a well-defined positive solution with  $p(0) = 0$ . This proves exactness for the stated relaxed problem. The largest uniformly safe scale  $k$  likewise follows from the admissible envelope  $v = \min(1, e^y - 1)$ , which attains  $\inf_y y/v(y) = k$ . These two extremizing envelopes need not be one fixed smooth scalar profile. No stronger minimax conclusion is used.  $\square$

## B.1 A globally smooth construction without velocity monotonicity

For  $A > 4$  define, on the whole metric line,

$$P_A(y) = \frac{1 + 2e^{-A} \cosh y}{1 + 2e^{-A}}, \quad X_A(y) = \int_0^y \frac{du}{P_A(u)}, \quad Z_A(y) = \int_0^y P_A(u) du.$$

The map  $X_A$  is a smooth increasing bijection from  $\mathbb{R}$  onto a finite interval  $(-b_A, b_A)$ :  $P_A(y) \geq e^{y-A}/(1 + 2e^{-A})$  for  $y \geq 0$ , and  $P_A$  is even. Define  $f_A$  by  $f'_A(X_A(y)) = Z_A(y)$  and  $f_A(0) = 0$ . All these functions are smooth, so no piecewise smoothing or boundary extension is implicit. Differentiation gives

$$f''_A(X_A(y)) = P_A(y)^2, \quad \frac{|f'''_A|}{(f''_A)^{3/2}} = 2 \frac{|P'_A|}{P_A} < 2, \quad V_A(y) := \frac{f'_A}{\sqrt{f''_A}} = \frac{y + 2e^{-A} \sinh y}{1 + 2e^{-A} \cosh y}. \quad (\text{B.4})$$

In particular  $y$  is its exact metric coordinate, and 0 is its analytic center. Since  $V_A(y) \rightarrow \pm 1$  as  $y \rightarrow \pm\infty$ , the identity  $\frac{d}{dy} f_A(X_A(y)) = V_A(y)$  proves barrier blow-up at both endpoints. Also  $Z_A(y) \rightarrow \pm\infty$ , so the gradient range is all of  $\mathbb{R}$ .

The exact scalar parameter is  $\nu_A = \sup_y V_A(y)^2$ . For  $0 \leq y \leq A$ ,  $V_A(y) \leq y + 1 \leq A + 1$ . For  $y = A + u \geq A$ , use  $2e^{-A} \sinh y \leq 2e^{-A} \cosh y$  and  $2e^{-A} \cosh y \geq e^u$  to get

$$V_A(y) \leq 1 + \frac{A + u - 1}{1 + e^u} \leq A + 1;$$

the final inequality follows from  $u - 1 \leq Ae^u$  for  $A > 4$ . Oddness handles negative  $y$ . At  $y = A/2$ , the denominator in (B.4) is  $1 + e^{-A/2} + e^{-3A/2} < 3/2$ , so  $V_A(A/2) > A/3$ . Thus  $A^2/9 < \nu_A \leq (A + 1)^2$ . The velocity is not monotone: it exceeds one at  $A/2$  but tends to one at the endpoint.

Fix  $r$  and  $0 < \delta < 1/2$ . Uniformly for  $y \in [\delta A, (1 - \delta)A]$  as  $A \rightarrow \infty$ ,  $P_A(y) = 1 + o(1)$  and  $Z_A(y) = y + o(1)$ . Consequently the logarithmic central-parameter width of this metric interval is

$$\log Z_A((1 - \delta)A) - \log Z_A(\delta A) = \log \frac{1 - \delta}{\delta} + o(1).$$



Choose  $B > \log((1-\delta)/\delta)$  and weights  $\log w_i = B(r-i)$ . Terminate at  $s_1 = \log Z_A((1-\delta)A)$ . The  $r$  translated windows are disjoint for large  $A$ ; each advances one metric coordinate by  $(1-2\delta)A$ . Therefore their central arc has length at least  $r(1-2\delta)A$ .

At termination,  $Z_A(y_i) = Z_A((1-\delta)A)e^{B(r-i)}$ . For  $y \geq \delta A$  and large  $A$ ,  $Z_A(y) \geq e^{y-A}/3$  and  $Z_A((1-\delta)A) < A$ . Thus  $y_i \leq A + \log(3A) + B(r-i)$ , giving endpoint distance at most  $\sqrt{r}[A + \log(3A) + Br]$ . Let  $A \rightarrow \infty$  with  $r, \delta, B$  fixed, then let  $\delta \downarrow 0$ . The ratio approaches  $\sqrt{r}$ , proving [proposition 7.2](#). In particular, no parameter-dependent comparison such as (1.2) is contradicted: the scalar parameter, and hence the product parameter  $r\nu_A$ , diverges in this construction.

## C Proofs for the concrete formulations

### C.1 Grouped paraboloids and arbitrary PSD column packing

For a grouped tangent  $(\dot{s}, \dot{w})$ , put  $\dot{q} = \dot{s} - 2\langle w, \dot{w} \rangle$ . Direct differentiation gives

$$D^2(-\log q)[(\dot{s}, \dot{w}), (\dot{s}, \dot{w})] = (\dot{q}/q)^2 + 2\|\dot{w}\|^2/q. \quad (\text{C.1})$$

For a packed block, an arbitrary affine tangent gives  $\dot{D} = \dot{S} - \dot{W}^T W - W^T \dot{W}$ , and

$$D^2(-\log \det D)[(\dot{S}, \dot{W}), (\dot{S}, \dot{W})] = \text{tr}((D^{-1}\dot{D})^2) + 2\text{tr}(D^{-1}\dot{W}^T \dot{W}). \quad (\text{C.2})$$

These are Hessians in affine  $(s, w)$  or  $(S, W)$  coordinates; treating  $D$  as an independent affine variable when taking the second derivative would omit the second term. With  $c_\ell$  columns in block  $\ell$ , trace Cauchy–Schwarz gives

$$\|\dot{z}\|_F^2 \geq \sum_\ell \frac{1}{c_\ell} \left( \frac{d}{dt} \log \det D_\ell \right)^2. \quad (\text{C.3})$$

Its scalar version is the simultaneous contraction onto every  $\log q$ . Both Hessians are positive definite on the free affine coordinates. Standard self-concordance follows by restricting the classical PSD barrier to the affine matrices  $\begin{pmatrix} I & W \\ W^T & S \end{pmatrix}$ , whose determinant is  $\det D$ ; the one-column case gives (C.1). The derivative  $DF = -\sum_\ell \text{tr}(D_\ell^{-1}\dot{D}_\ell)$  and (C.3) imply  $|DF[\dot{z}]| \leq \sqrt{H}\|\dot{z}\|_F$ , where  $H = \sum_\ell c_\ell = kh$ . This proves the parameter upper bound  $H$  without charging the fixed identity blocks.

For a feasible target write  $\delta_a = 1 - \langle c_a, w_a \rangle$ . The diagonal residuals  $d_\gamma = (D_{\ell(\gamma)})_{\gamma\gamma}$  satisfy

$$\sum_{\gamma \text{ from } a} d_\gamma = 1 - \|w_a\|^2 \leq 1 - \langle c_a, w_a \rangle^2 \leq 2\delta_a.$$

The same identity holds for grouped  $q_{aG}$ . If the target error is at most  $\varepsilon$ , then  $\sum_a \delta_a \leq \varepsilon$  and

$$\prod_\ell \det D_\ell \leq \prod_\gamma d_\gamma \leq \prod_a (2\delta_a/h)^h \leq (2\varepsilon/H)^H. \quad (\text{C.4})$$

The first inequality is Hadamard’s inequality and permits every completion variable to vary; it is not a diagonal-fiber assumption. Integrating (C.3) and using weighted Cauchy–Schwarz shows that every joining curve has length at least  $H^{-1/2}$  times the absolute change in the sum of log determinants. If the reference has  $D_\ell = q_0 I$ , (C.4) gives (10.8).

For completeness, these are indeed the claimed centers. At fixed  $W$  and fixed residual diagonals, Hadamard’s inequality minimizes the barrier precisely when every residual  $D_\ell$  is diagonal. Partial

minimization over the unrestricted off-diagonal entries of  $S$  therefore leaves the grouped problem. Its equations in  $s_{aG}$  give one common residual  $q$  per ball, and its equations in  $w$  give  $w_{aG} = (\tau q/2)c_{aG}$ . The source equation becomes  $hq + (\tau q/2)^2 = 1$ , proving (10.7). Positive definite Hessians make these stationary points the unique centers. Differentiated central stationarity on the affine slice yields  $H_\tau z'(\tau) = \nabla \ell$  and hence

$$\left\| \frac{dz}{d \log \tau} \right\|^2 = \tau^2 \frac{d}{d\tau} \ell(z(\tau)) = k\tau^2 r'(\tau) = H \left( 1 - \frac{h}{\sqrt{h^2 + \tau^2}} \right).$$

At a center,  $DF = \tau d\ell$  on the slice, so the same expression is the squared dual gradient norm. Its limit  $H$  proves the exact parameter. Changing variables from  $\tau$  to  $r$  gives the length element  $\sqrt{H}\rho'(r) dr$ , proving (10.12).

## C.2 Linked norm-tree centers and distance

The positive-sheet domain (10.9) is a convex affine restriction of a product of Lorentz cones. Its Hessian is positive definite: any nonzero free direction changes at least one cone block, whose Hessian is positive definite. Restriction and addition preserve standard self-concordance. Telescoping cancels every nonroot internal square and gives, for each source,

$$\sum_v q_{av} = 1 - \|w_a\|^2. \quad (\text{C.5})$$

The derivative in a nonroot axial variable is  $-2t_{av}/q_{av} + 2t_{av}/q_{a,\text{parent}(v)}$ . Since  $t_{av} > 0$ , central stationarity forces all residuals in a tree to agree. Leaf stationarity gives  $w_a = rc_a$ , and (C.5) becomes  $bq + r^2 = 1$ , with  $r = \tau q/2$ . The positive internal coordinates are uniquely  $t_{av} = (n_v q + r^2 C_{av})^{1/2}$ ; their root value is one. These equations prove feasibility and stationarity, and strict convexity proves uniqueness. They give the analytic center  $w = 0$ ,  $t_{av}^2 = n_v/b$  and the path/speed formulas asserted in the main text.

For each Lorentz block, the parameter-two gradient inequality gives  $(d \log q_{av})^2 \leq 2D^2(-\log q_{av})$  after restriction. Therefore the total metric controls the Euclidean log-residual metric with factor  $1/\sqrt{2}$ . More sharply, with the exact restricted parameter  $\nu$  proved below, the barrier itself is  $\sqrt{\nu}$ -Lipschitz in metric length. Telescoping and the same source AM–GM calculation as (C.4) give

$$\prod_{a,v} q_{av}(y) \leq (2\varepsilon/L)^L, \quad F(y) - F(\bar{z}) \geq L \log(k/(2\varepsilon)).$$

This proves (10.11). Differentiated stationarity gives the same speed as for grouped balls with  $h = b, H = L$ , so the same scalar integration proves (10.12). Thus no asymptotic regularity of boundary KKT equations is needed for the matched route.

## C.3 Exact root-slice parameter

We prove (10.10) for one tree. Before fixing the root  $t$ , let  $\Phi = -\sum_v \log q_v$  on the linked cone and  $H = \Phi''$ . Its logarithmic degree is  $2b$ ; hence  $H z = -\Phi'(z)$  and  $\|\Phi'\|_{H^{-1}}^2 = 2b$ . Let  $e_t$  be the root-coordinate covector. Orthogonal projection of the gradient onto  $\{h : e_t^T h = 0\}$  gives the exact restricted norm

$$\|DF\|_{F,*}^2 = 2b - \frac{t^2}{e_t^T H^{-1} e_t}, \quad \nu_T = 2b - \frac{1}{\sup_z \beta_T(z)}, \quad \beta_T(z) = \frac{e_t^T H^{-1} e_t}{t^2}. \quad (\text{C.6})$$

Dikin containment and the halfspace  $t > 0$  imply  $e_t^T H^{-1} e_t \leq t^2$ , so  $\beta_T \leq 1$ . Indeed, if the inequality failed, a vector of local norm less than one could move the root coordinate to zero.

If a root child is a free leaf, set  $t = 1$  and that leaf to  $\sqrt{1 - \delta}$ . Set other root leaves to zero and scale every internal child subtree from a fixed interior point by  $\delta^2$ . The root residual is  $q = \delta - O(\delta^4) > 0$ . For the ambient trial direction  $h_t = 1$ ,  $h_{\text{leaf}} = 1$ , and every other component zero, descendant blocks contribute zero, while

$$D^2 \Phi[h, h] = \frac{\{2(1 - \sqrt{1 - \delta})\}^2}{q^2} \rightarrow 1.$$

Here the root residual's second derivative in that direction is zero. The variational identity  $(e_t^T H^{-1} e_t)^{-1} = \min_{h_t=1} h^T H h$  now gives  $\sup \beta_T = 1$ . This includes a one-node tree without descendants. It proves  $\nu_T = 2b - 1$  with an explicit feasible limiting sequence.

Suppose instead that all root children are internal, with positive axial coordinates  $a_1, \dots, a_c$ . Eliminate all their strict descendants from the Hessian. The effective axial curvature  $h_i$  of child subtree  $i$  is at least  $a_i^{-2}$  by the same Dikin halfspace argument on that subtree. Increasing an axial curvature increases the root Schur complement, so a lower bound is obtained from

$$-\log(t^2 - \sum_i a_i^2) - \sum_i \log a_i.$$

Normalize  $t = 1$ , put  $r = \sum_i a_i^2$ ,  $q = 1 - r$ , and  $Z = \sum_i a_i^4 / (q + 2a_i^2)$ . Its child Hessian is a diagonal matrix with entries  $2/q + a_i^{-2}$  plus  $4aa^T/q^2$ ; the root cross row is  $-4a^T/q^2$ , and the root entry is  $2(1 + r)/q^2$ . The rank-one inverse formula therefore gives the root Schur complement

$$S = \frac{2(1 + r) - 16Z/(q + 4Z)}{q^2}. \quad (\text{C.7})$$

Since  $x/(q + 2x)$  increases in  $x \geq 0$ ,  $\sum_i a_i^4 / (q + 2a_i^2) \leq r^2 / (q + 2r) = r^2 / (1 + r)$ . The inequality  $S \geq 2$  is equivalent to  $r(3 - r) \geq 4(2 - r)Z$ ; replacing  $Z$  by this upper bound leaves  $3r(1 - r)^2 / (1 + r) \geq 0$ . Thus  $\beta_T \leq 1/2$ . Conversely, scale all child subtrees by  $\delta \downarrow 0$ . The trial direction  $h_t = 1$ , with all other variables fixed, has norm squared  $2(1 + r)/(1 - r)^2 \rightarrow 2$ . Together with the lower bound on the minimizing Schur complement this proves  $\sup \beta_T = 1/2$  and  $\nu_T = 2b - 2$ . For independent trees, gradient norm squares and their suprema add, proving the product parameter used in (10.11).

## C.4 Heterogeneous source counts and objective weights

A single weighted version records the dependence on source sizes for all three formulations. Let source  $a$  have  $m_a$  groups or internal nodes, let  $M_0 = \sum_a m_a$ ,  $p_a = m_a/M_0$ , and maximize  $\sum_a \lambda_a \langle c_a, w_a \rangle$  with every  $\lambda_a > 0$ . Define the scale

$$\Delta_{\text{eff}} = \prod_a (\lambda_a / p_a)^{p_a}. \quad (\text{C.8})$$

At the separate analytic centers all source residuals are  $1/m_a$ . For an accurate target, put  $e_a = \lambda_a(1 - \langle c_a, w_a \rangle)$ , so  $\sum_a e_a \leq \varepsilon$ . Telescoping, or the residual diagonal identities and Hadamard's inequality, show that the ratio of terminal to central residual products is at most  $\prod_a (2e_a / \lambda_a)^{m_a}$ . This product is maximized at  $e_a = \varepsilon p_a$ , giving

$$F(y) - F(\bar{z}) \geq M_0 \log(\Delta_{\text{eff}} / (2\varepsilon)).$$

Consequently the entire accurate set is at distance at least

$$\frac{M_0}{\sqrt{\nu}} [\log(\Delta_{\text{eff}} / (2\varepsilon))]_+, \quad \nu = \begin{cases} M_0, & \text{grouped or packed,} \\ \sum_a \nu_{T_a}, & \text{norm trees.} \end{cases} \quad (\text{C.9})$$

The grouped/packed parameter remains exact by applying the center calculation source by source, with parameter  $\tau\lambda_a$ . For equal weights  $\lambda_a = \lambda > 0$ ,  $\Delta_{\text{eff}} = \lambda \exp(-\sum_a p_a \log p_a)$ ; weighted AM–GM gives  $\Delta_{\text{eff}} \leq \sum_a \lambda_a$  with equality precisely when  $\lambda_a$  is proportional to  $m_a$ . Thus the scale, as well as the square-root count, matters for movement.

## References

- [1] Xavier Allamigeon, Pascal Benchimol, Stéphane Gaubert, and Michael Joswig. Log-barrier interior point methods are not strongly polynomial. *SIAM Journal on Applied Algebra and Geometry*, 2(1):140–178, 2018. doi: 10.1137/17M1142132.
- [2] Xavier Allamigeon, Stéphane Gaubert, and Nicolas Vandame. No self-concordant barrier interior point method is strongly polynomial. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 515–528. Association for Computing Machinery, 2022. doi: 10.1145/3519935.3519997.
- [3] Xavier Allamigeon, Daniel Dadush, Georg Loho, Bento Natura, and László A. Végh. Interior point methods are not worse than Simplex. *SIAM Journal on Computing*, 54(5):FOCS22–178–FOCS22–264, 2025. doi: 10.1137/23M1554588.
- [4] Michel Baes. Convexity and differentiability properties of spectral functions and spectral mappings on Euclidean Jordan algebras. *Linear Algebra and its Applications*, 422(2–3):664–700, 2007. doi: 10.1016/j.laa.2006.11.025.
- [5] Sébastien Bubeck and Ronen Eldan. The entropic barrier: Exponential families, log-concave geometry, and self-concordance. *Mathematics of Operations Research*, 44(1):264–276, 2019. doi: 10.1287/moor.2017.0923.
- [6] Jordi Castro and Jordi Cuesta. Quadratic regularizations in an interior-point method for primal block-angular problems. *Mathematical Programming*, 130(2):415–445, 2011. doi: 10.1007/s10107-010-0341-2.
- [7] Sinho Chewi. The entropic barrier is  $n$ -self-concordant. In Ronen Eldan, Bo’az Klartag, Alexander Litvak, and Emanuel Milman, editors, *Geometric Aspects of Functional Analysis*, volume 2327 of *Lecture Notes in Mathematics*, pages 209–222. Springer, Cham, 2023. doi: 10.1007/978-3-031-26300-2\_6.
- [8] Daniel Dadush, Haoyuan Ma, Bento Natura, and László A. Végh. Trust region interior point methods: Optimal  $\ell_2$ - and faster wide-neighborhood path following. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, pages 755–766. Association for Computing Machinery, June 2026. doi: 10.1145/3798129.3800790.
- [9] Antoine Deza, Eissa Nematollahi, and Tamás Terlaky. How good are interior point methods? Klee–Minty cubes tighten iteration-complexity bounds. *Mathematical Programming*, 113:1–14, 2008. doi: 10.1007/s10107-006-0044-x.
- [10] Jacques Faraut and Adam Korányi. *Analysis on Symmetric Cones*. Oxford Mathematical Monographs. Clarendon Press, Oxford, 1994. doi: 10.1093/oso/9780198534778.001.0001.
- [11] João Gouveia, Pablo A. Parrilo, and Rekha R. Thomas. Lifts of convex sets and cone factorizations. *Mathematics of Operations Research*, 38(2):248–264, 2013. doi: 10.1287/moor.1120.0575.

- [12] Raphael A. Hauser and Osman Güler. Self-scaled barrier functions on symmetric cones and their classification. *Foundations of Computational Mathematics*, 2(2):121–143, 2002. doi: 10.1007/s102080010022.
- [13] Roland Hildebrand. A lower bound on the barrier parameter of barriers for convex cones. *Mathematical Programming*, 142:311–329, 2013. doi: 10.1007/s10107-012-0576-1.
- [14] Hiroshi Hirai, Harold Nieuwboer, and Michael Walter. Interior-point methods on manifolds: Theory and applications. *Foundations of Computational Mathematics*, 2026. doi: 10.1007/s10208-026-09756-8. Published online.
- [15] Yin Tat Lee and Man-Chung Yue. Universal barrier is  $n$ -self-concordant. *Mathematics of Operations Research*, 46(3):1129–1148, 2021. doi: 10.1287/moor.2020.1113.
- [16] Lucas Lévy, Jean-Lou Valeau, Arya Akhavan, and Patrick Rebeschini. Self-concordant perturbations for linear bandits, 2026. [arXiv:2510.24187v3](https://arxiv.org/abs/2510.24187v3).
- [17] Adrian S. Lewis and Hristo S. Sendov. Twice differentiable spectral functions. *SIAM Journal on Matrix Analysis and Applications*, 23(2):368–386, 2001. doi: 10.1137/S089547980036838X.
- [18] George G. Lorentz. On the theory of spaces  $\Lambda$ . *Pacific Journal of Mathematics*, 1(3):411–429, 1951. doi: 10.2140/pjm.1951.1.411.
- [19] Yurii Nesterov and Arkadi Nemirovski. Primal central paths and Riemannian distances for convex sets. *Foundations of Computational Mathematics*, 8(5):533–560, 2008. doi: 10.1007/s10208-007-9019-4.
- [20] Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*, volume 13 of *SIAM Studies in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, 1994. doi: 10.1137/1.9781611970791.
- [21] Yurii E. Nesterov and Michael J. Todd. On the Riemannian geometry defined by self-concordant barriers and interior-point methods. *Foundations of Computational Mathematics*, 2(4):333–361, 2002. doi: 10.1007/s102080010032.
- [22] Atsumi Ohara, Hideyuki Ishi, and Takashi Tsuchiya. Doubly autoparallel structure and curvature integrals: Applications to iteration complexity for solving convex programs. *Information Geometry*, 7:555–586, 2024. doi: 10.1007/s41884-023-00116-x.
- [23] Erik Alex Papa Quiroz and Paulo Roberto Oliveira. New self-concordant barrier for the hypercube. *Journal of Optimization Theory and Applications*, 135(3):475–490, 2007. doi: 10.1007/s10957-007-9220-2.
- [24] Frank Permenter. A geodesic interior-point method for linear optimization over symmetric cones. *SIAM Journal on Optimization*, 33(2):1006–1034, 2023. doi: 10.1137/20M1385019.